

Program overview

19-May-2025 13:39

Year	2024/2025
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Data Science and Artificial Intelligence Technology

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master DSAIT 2024			
Master Data Science and Artificial Intelligence Technology 2024			
Core courses 2024			
DSAIT4000	Data Management and Engineering	5	
DSAIT4005	Machine and Deep Learning	5	
DSAIT4010	Probabilistic AI and Reasoning	5	
DSAIT4015	Software Engineering and Testing for AI Systems	5	
DSAIT4500	Responsible Data Science and AI Engineering	5	
Themes 2024			
Advanced Machine Learning 2024			
DSAIT4020	Elements of Statistical Learning	5	
DSAIT4025	Alternative Learning Strategies	5	
DSAIT4030	Generative Modelling	5	
Bioinformatics 2024			
DSAIT4035	Algorithms for Sequence-based Bioinformatics	5	
DSAIT4040	Algorithms for Network-based Bioinformatics	5	
DSAIT4045	Machine Learning in Bioinformatics	5	
Data and Information Management 2024			
CS4525	Web-Scale Data Management	5	
DSAIT4050	Information Retrieval	5	
DSAIT4055	Web Science and Engineering	5	
Human-Centred AI 2024			
DSAIT4060	Human-centered Machine Perception	5	
DSAIT4065	Conversational Agents	5	
DSAIT4070	Designing Human-Centred AI Systems	5	
Interactive Visual Analysis 2024			
DSAIT4075	Data Visualization	5	
DSAIT4080	3D Visualization	5	
DSAIT4085	High-Performance Analysis Systems	5	
Language and Speech Technology 2024			
DSAIT4090	Natural Language Processing	5	
DSAIT4095	Inclusive Speech Technology	5	
DSAIT4100	Natural Language Processing for Society	5	
Optimisation & Reasoning 2024			
CS4205	Evolutionary Algorithms	5	
CS4530	Modelling and Problem Solving	5	
CS4535	Constraint Solving	5	
Probabilistic Decision Making 2024			
DSAIT4105	Probabilistic Models and Inference	5	
DSAIT4110	Sequential Decision Making	5	
DSAIT4115	Deep Reinforcement Learning	5	
Software Engineering for DS and AI 2024			
CS4295	Release Engineering for Machine Learning Applications	5	
CS4570	Machine Learning for Software Engineering	5	
CS4575	Sustainable Software Engineering	5	
Visual Computing 2024			
DSAIT4120	Applied Image Processing	5	
DSAIT4125	Computer Vision	5	
DSAIT4130	3D Visual Computing	5	
Research courses 2024			
CESE4065	Advanced Practical I.o.T. and Seminar	5	
DSAIT4200	Empirical Research of Computational Solutions	5	
DSAIT4205	Fundamental Research in Machine and Deep Learning	5	
DSAIT4210	Research in Web Data and Information Management	5	
DSAIT4215	Research in Visual Computing and Interactive Visual Analysis	5	
DSAIT4220	Research in Intelligent Decision Making	5	
DSAIT4225	Research in Bioinformatics	5	
DSAIT4230	Research in Social Signal Processing and Affective Computing	5	
DSAIT4235	Research Seminar on Multimedia Computing and Systems	5	
Deepening/broadening Electives 2024			

Interdisciplinary projects			
IFEEMCS520201	Advanced Interdisciplinary Artificial Intelligence Project	15	
IFM4040	Joint Interdisciplinary Project	15	
TPM423A	Technology Venture Development	15	
Deepening electives			
DSAIT4300	Formal Methods for Machine Learning	5	
DSAIT4305	Machine Learning for Graph Data	5	
DSAIT4310	Modeling and Data Analysis and Complex Networks	5	
DSAIT4315	Performance Analysis	5	
DSAIT4320	Building Serious Games	5	
DSAIT4325	Socio-Cognitive Engineering	5	
Broadening electives			
TPM027A	Cyber Risk Management	5	
TPM401A	Technology Entrepreneurship and Innovation	5	
TPM404B	Technology Entrepreneurship and Internationalization	5	
TPM409A	Digital technology entrepreneurship and management	5	
TPM411A	Idea to Start-up IT & AI	5	
Thesis Project 2024			
DSAIT5000	Thesis Project	45	

Year

Organization

Education

2024/2025

Electrical Engineering, Mathematics and Computer Science

Master Data Science and Artificial Intelligence Technology

Master DSAIT 2024

Core courses 2024	
Introduction 1	To form a solid understanding of the field of DSAIT, you will develop general competence on advanced theoretic concepts of Data Management and Engineering, Machine and Deep Learning, Software Engineering and Testing for AI Systems, Probabilistic AI and Reasoning, and Responsible Data Science and AI Engineering. These are reflected in the common core of the programme.
Introduction 2	Common core courses: DSAIT4000 Data Management and Engineering DSAIT4005 Machine and Deep Learning DSAIT4010 Probabilistic AI and Reasoning DSAIT4015 Software Engineering and Testing for AI Systems DSAIT4500 Responsible Data Science and AI Engineering

DSAIT4000	Data Management and Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. R. Hai	
Instructor	Dr.ir. U.K. Gadiraju	
Instructor	J. Yang	
Instructor	T. Höllt	
Instructor	Dr. A. Katsifodimos	
Instructor	Dr. A. Anand	
Instructor	Dr. M.S. Pera	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides a comprehensive introduction into a large variety of core topics related to data, data management, and data engineering. The quality of data and its management plays a central role in AI and data-driven systems, and has a direct impact on a system's effectiveness, efficiency, safety and reliability. Therefore, the following topics are being covered: - The role of data in AI systems - Data Models and modelling data - Discovering and creating large datasets - Data quality and improving data quality - Visually exploring data - Integrating datasets - Analyzing data with data mining techniques - Visualizing data and analysis results - Storing and retrieving data - Distributing data - Impact of inadequate data and data handling on subsequent systems and decision making	
Study Goals	At the end of this course, the student is able to: 1. Understand the role of data in data-driven/AI applications and commonly encountered data models and data-related problems 2. Apply core methodologies and techniques for storing and retrieving data, also in a distributed fashion 3. Apply core methodologies and techniques for obtaining, cleaning and integrating data as part of a larger data-processing pipeline 4. Apply core methodologies and techniques for visualizing and visual exploration of data 5. Apply core methodologies and techniques for analysing and mining datasets 6. Analyse a given Data Engineering problem and create a suitable solution and evaluate its effectiveness and efficiency	
Education Method	In-person lectures with additional labs/practical projects (with extra online instructions/materials)	
Literature and Study Materials	Custom-made slides and selected research papers	
Assessment	The final grade of the course consist of the following components: - Written Exam (weighting 60%) - Group Assignments: small-group projects focusing on practical skills and enabling key skills / professional skills (weighting 40%) Final grade calculation = 0.6 * Written Exam + 0.4 * Group Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4005	Machine and Deep Learning	5
Responsible Instructor	Dr. D.M.J. Tax	
Responsible Instructor	Dr. J.C. van Gemert	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The range of problems that a computer scientist may encounter is very broad, from the very abstract (to determine if some problem can be solved by a computer) to very specific (to implement an electronic health record system). Due to the complexity and noisiness of the real world, a modern computer scientist has to rely more and more on the analysis and models of data. Machine Learning, and a recently successful extension of it, Deep Learning, deals with the analysis of data and the development of models. To be able to apply these algorithms, the computer scientist needs to have a good understanding of the foundations of Machine Learning. This course provides a recapitulation of (un)supervised learning, classification, decision theory and overfitting. It introduces some classical statistical learning classifiers and discusses their complexity, and ways to adapt their complexity (regularisation). The course focuses further on Deep Learning, including backpropagation, Optimization, Convolutional Neural Networks, Recurrent Neural Networks, self-attention and Transformers. Finally, the course discusses biases in datasets, and the design and analysis of Machine Learning experiments. This makes the student aware how the Machine and Deep Learning models can be applied in a responsible manner to real world problems.	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and assumptions in the fields of Machine and Deep Learning, like the classical supervised learning, generalization, overfitting, model selection, and specialized architectures such as CNNs, RNNs, Transformers, etc. 2. Reflect on the strengths and limitations of classical and modern Machine and Deep Learning concepts 3. Apply the covered techniques, with reasons about their effectiveness 4. Implement key Machine Learning and Deep Learning techniques	
Education Method	In-person lectures with additional labs/practical projects	
Literature and Study Materials	- Bishop, "Pattern Recognition and Machine Learning" - Goodfellow et al., "Deep Learning"	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 100%) Final grade calculation = 1 * Digital Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4010	Probabilistic AI and Reasoning	5
Responsible Instructor	Dr. N. Yorke-Smith	
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Prof.dr. M.M. de Weerd	
Instructor	S. Dumani	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Artificial intelligence and data science deal with a variety of complex problems ranging from face recognition to inventory management and controlling humanoid robots. For such tasks it is not immediately clear how to manually program good solutions. Instead, many approaches to successfully dealing with such complex problems are based on modelling the problem in a formal framework (such as logic, a graphical model, or Bayesian network), and applying corresponding reasoning or optimization techniques to find solutions. This course aims to introduce a number of such frameworks and how they can be used to model real-world problems and often also the uncertainties that arise in them. This includes discussing relations between these models and their associated methods like search, inference, learning and optimization. The specific topics include logical modelling, constraint satisfaction problems, graphical models, Bayesian networks, as well as temporal reasoning and learning via Markov decision processes and reinforcement learning.	
Study Goals	At the end of this course, the student is able to: 1. Comprehensively summarize the covered topics (what is AI?, modelling, graphical models and reasoning, bayesian networks, MDPs, reinforcement learning, etc.) and relevant aspects. 2. Explain the relations between search, inference, learning and optimization, as well as explain the main concepts and techniques. 3. Appraise the role of modelling, reasoning and learning in AI, based on examples of specific modelling approaches. 4. Apply covered modelling approaches (such as graphical models, bayesian networks, and markov decision processes) to practical real-world problems. 5. Reason about the effectiveness of certain solution (search/learning/inference/etc.) approaches for specific types of problems.	
Education Method	Lectures, homework exercises	
Literature and Study Materials	* Russell & Norvig. Artificial Intelligence: A Modern Approach, 4th Global ed., 2021 * provided materials and readings	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) Final grade calculation = 1 * Written Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Special Information	Course email address: dsait-pair-cs-ewi@tudelft.nl	
Tags	Artificial intelligence	

DSAIT4015	Software Engineering and Testing for AI Systems	5
Responsible Instructor	Dr.ir. C.C.S. Liem	
Responsible Instructor	Dr. A. Panichella	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	How do you know and assess whether an AI-intensive system is working as intended? On the one hand, quality assurance for such systems can be considered to be a software testing challenge; in parallel, the machine learning components within such systems are normally validated according to machine learning-specific evaluation methodology. In this course, you will learn more about these two complementary takes to evaluation and quality assurance, while learning about (and getting hands-on experience in) state-of-the-art advances in combining these two takes. Course content taught includes: - Basics of testing (assertions, white-box vs. black-box testing) - Basics of machine learning evaluation (validation & validity) - Quality assurance at different scopes in AI-intensive systems (models, components, systems, cyber-physical systems) - Adversarial attacks - Software testing techniques (e.g. fuzzing, metamorphic testing, combinatorial testing, differential testing) - Data labels (ground truth/oracle) validation	
Study Goals	At the end of this course, the student is able to: 1. Apply state-of-the-art techniques to test and validate machine learning models and training routines 2. Improve robustness considerations in AI-intensive systems upon analysis of their current workings 3. Implement techniques to generate application- and context-specific adversarial attacks 4. Connect the oracle problem in software testing to questions of experimental validity in ground truth 5. Complement classical evaluation reporting of machine learning models with insights obtained through testing	
Education Method	Lectures, project	
Literature and Study Materials	Provided by the lecturers in the form of relevant papers and grey literature (on real-world case studies)	
Assessment	The final grade of the course consists of the following components: - Group Report (weighting 70%) - Digital Exam (weighting 30%) Final grade calculation = 0.7 * Group Report + 0.3 * Digital Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Report: Repair opportunity - Digital Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4500	Responsible Data Science and AI Engineering	5
Responsible Instructor	Dr.ir. C.C.S. Liem	
Responsible Instructor	L. Cavalcante Siebert	
Instructor	Dr. A. Anand	
Instructor	Dr. M.S. Pera	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The goal of this course is for students to further develop a responsible attitude that will contribute to the role they play in their future careers. Students will learn to make and reflect on choices related to Responsible DSAIT Engineering with an ethical and sustainable stance and act with integrity. Students will engage with social, organizational, professional, and technical aspects of developing DSAIT, from both systems and human-centric perspectives. Topics discussed include human agency and oversight, algorithmic robustness and safety, privacy and data governance, transparency, non-discrimination, fairness, inclusivity, accessibility, and accountability. The course will run in the first year of the master, in quarter 3. Responsible DSAIT Engineering-related content in courses in quarter 1 and 2 will be a means for students to start reflecting on their professional roles and on their work in the different courses and projects they undertake. No deliverables are expected not required in quarter 1 and 2. In this course, students will be presented with organizational, social and technical approaches towards designing and developing Ethical and Responsible DSAIT, integrity and ethics in DSAIT and engineering, discuss and engage with colleagues on their duties and roles in their future careers, connect different perspectives related to ethical and responsible DSAIT as they think of new problems, and prepare their final presentations.	
Study Goals	At the end of this course, the student is able to: 1. Reflect on your role, responsibilities, possible dilemmas and trade-offs in developing DSAIT, both as a professional and as a team member. 2. Analyse DSAIT approaches and principles towards addressing societal and ethical issues, considering the perspectives of multiple stakeholders. 3. Contribute to appropriate responsible design, development and impact assessments on DSAIT work. 4. Integrate concepts related to trustworthy DSAIT and integrity when presenting and justifying your choices to address matters of responsibility connected to DSAIT.	
Education Method	Most lectures with 45min content + 45 min discussions (guided by cases). In addition, selected content and guest lectures.	
Literature and Study Materials	Slides, real-life case scenarios, scientific articles, and impact assessment methods.	
Assessment	The final grade of the course consists of the following components: - Group Report: group case study analysis and presentation (P/F) - Individual Presentation: individual interview (P/F) Both components need to be passed, to receive a pass for the course. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Report: Repair opportunity - Individual Presentation: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. M.S. Pera	

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Themes 2024	
Introduction 1	Through our challenging themes, you obtain advanced knowledge and skills in subject areas of Data Science and AI Technology through inspiring themes, allowing you to work on specific types of challenges you might encounter as a professional. With the flexibility to select two themes, you have the freedom to create a programme that is tailored to your individual strengths and area of interest. Each theme has 3 courses in the 2nd, 3rd and 4th quarters of the 1st year.
Introduction 2	Themes: - Advanced Machine Learning - Bioinformatics - Data Information Management - Human-centred AI - Interactive Visual Analysis - Language and speech technology - Optimisation and Reasoning - Probabilistic Decision Making - Software Engineering for Data Science and AI - Visual Computing

Advanced Machine Learning 2024	
Introduction 1	In Advanced Machine Learning the fundamentals and the current state of the research field of Machine Learning is taught. The idea that a machine (computer) can learn from data has intrigued many researchers throughout the ages. Therefore, the field of machine Learning is influenced by ideas and techniques from statistics, decision theory, optimization, physics, and many other disciplines. To form a coherent picture of the field in this theme, the courses are ordered from the more classical statistical learning to the most modern alternative learning strategies. The theme covers techniques that learn from data: - Classical statistical learning, including supervised and unsupervised learning, classification, regression, clustering, feature extraction, complexity, regularization - Alternative learning strategies, like online learning, meta-learning, semi-supervised learning, causal inference, etc. - Generative modelling, including fully-observed, latent-variable, energy-based and score-based methods as well as the inference methods In the application of these techniques to practical problems, unavoidably they have to be tuned or adjusted. The focus is therefore not to only apply these techniques, but to gain a fundamental understanding of these techniques and then propose adjustments or improvements. The student should therefore know their strengths and limitations, and should be able to adapt or extend them for the application at hand.
Introduction 2	Courses: Q2: DSAIT4020 Elements pf Statistical Learning Q3: DSAIT4025 Alternative Learning Strategies Q4: DAIT4030 Generative Modeling

DSAIT4020	Elements of Statistical Learning	5
Responsible Instructor	Dr. D.M.J. Tax	
Responsible Instructor	T.J. Viering	
Instructor	Dr. J. Sun	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2 Summer Holidays	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: Linear Algebra, Probability Theory and Statistics, Machine and Deep Learning core course	
Course Contents	This course covers all the basic concepts of Statistical Learning, focusing on the classical techniques of Machine Learning before the era of Deep Learning. These concepts include classification, (ridge) regression, (hierarchical) clustering, feature reduction and extraction, model selection and bootstrapping, fairness in ML. The emphasis is on the concepts rather than the mathematical details.	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and assumptions in the field of Statistical Learning, like the classical supervised learning, but also regression, clustering and dimensionality reduction 2. Reflect on the strengths and limitations of classical and modern Statistical Learning techniques. 3. Apply the covered learning techniques to analyse a dataset and reason about their effectiveness	
Education Method	There are lectures introducing the topics, a lab course and a final project where a dataset is analysed.	
Literature and Study Materials	"Elements of Statistical Learning" Hastie et al.	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 60%) - Group Report: project report on analysis of a dataset (weighting 40%) Final grade calculation = $0.6 * \text{Digital Exam} + 0.4 * \text{Group Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.	
	In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Group Report: Repair opportunity	
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DSAIT4025	Alternative Learning Strategies	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Responsible Instructor	T.J. Viering	
Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Instructor	N. Tömen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: Machine and Deep Learning core course, Elements of Statistical Learning course	
Course Contents	Many applications of machine learning do not fit exactly into the classical supervised learning setting. This course covers machine learning scenarios and strategies beyond the classical setting, which are relevant to contemporary machine learning research and applications. Examples are causal, meta, adversarial, multiple instance and semi-supervised learning.	
Study Goals	At the end of this course, the student is able to: 1. Identify which setting beyond (un)supervised learning a problem belongs to 2. Explain the essential problem that (adversarial/semi-supervised/causal/meta-/active/online/...) learning addresses and how it differs from the classical ML setting 3. Evaluate methods that are proposed in these learning scenarios, both theoretically and empirically 4. Mathematically prove basic properties of methods in these scenarios 5. Apply common methods in these learning scenarios on real-world datasets 6. Develop machine learning techniques adapted for a particular learning task	
Education Method	Weekly lecture (2 hours) and tutorial session (2 hours), reading material (including scientific papers) and exercises. Parts of these exercises constitute the 3 formative assignments that have to be handed in, where students have to critically reflect on research papers, program small simulations, prove properties of methods and apply methods to real-world datasets	
Literature and Study Materials	We use recent monographs and papers that are relevant for each of the topics. For example, for the causal machine learning we use a mix of material from books and papers: - Hernán, Miguel A, and James M Robins. Causal Inference: What If. CRC Press, 2020 - Pearl, J., Glymour, M., & Jewell, N. P. (2016). Causal Inference in Statistics: A Primer. Wiley. - Künzle, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. Proceedings of the National Academy of Sciences, 116(10), 41564165. https://doi.org/10.1073/pnas.1804597116.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) - Individual Assignment: assignment theoretically and empirically evaluating and developing methods and critiquing a research paper (pass/fail) Final grade calculation = Written Exam Grade (if pass for all three individual assignments) A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4030	Generative Modelling	5
Responsible Instructor	Dr. C.A. Raman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: Linear Algebra, Probability and Statistics, Machine Learning Recommended: Probabilistic Graphical Models, Bayesian Statistics	
Course Contents	The course will primarily focus on generative modeling and inference techniques. Regarding modeling, we will begin by exploring the classification of generative models and conducting a comprehensive study of fully-observed models, including Flow-Based Models, Autoregressive models, Random Fields, Log-linear models, Markov Models, and more. We will then delve into the latent variable models such as probabilistic PCA, diffusion models, variational autoencoders and relevant reparameterization techniques. Additionally, we will cover energy-based models like restricted Boltzmann machines, deep belief networks, as well as score-based models such as Generative Adversarial Networks. On the inference side, the course will encompass various learning methods related to generative models. This includes maximum likelihood, Monte Carlo methods, exact methods, Laplace approximation, maximum a posteriori estimation, methods of moments, variational inference, and Markov Chain Monte Carlo. Furthermore, we will examine the practical applications that utilize generative modeling, particularly in the domains of Image Synthesis, Natural Language Processing (NLP), and multimodal learning. Through this course, students will gain a comprehensive understanding of generative modeling techniques and inference methods, contrast their strengths and weaknesses, and apply them to real-world tasks in various domains.	
Study Goals	At the end of this course, the student is able to: 1. Explain generative models and their inference techniques mathematically. 2. Contrast the strengths, weaknesses, and limitations of the different generative modeling techniques. 3. Evaluate the effectiveness and performance of self-implemented modeling techniques on real-world problems. 4. Identify real-world scenarios where they are the suitable tool of choice. 5. Appraise the role of deep generative modeling techniques in contemporary AI research through group discussions of relevant publications.	
Education Method	Lecture, project, and paper discussion.	
Literature and Study Materials	Tomczak, J. M. (2022). Deep generative modeling. Springer. Foster, D. (2022). Generative deep learning. O'Reilly Media, Inc.	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 70%) - Individual Report (weighting 30%) Final grade calculation = $0.7 * \text{Digital Exam} + 0.3 * \text{Individual Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Individual Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	S. Tan	

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Bioinformatics 2024

Introduction 1

The Bioinformatics theme focuses on data science and artificial intelligence methodology for the future of life sciences and healthcare. Recent technological developments enabling readout, synthesis, and editing at the molecular level are revolutionising the study of living organisms. With high-dimensional complex molecular data being generated at an unprecedented rate, across different levels of abstraction and data modalities, new generations of computational methods are needed to leverage the inherent potential of these data to make new discoveries about complex biological systems and realise exciting possibilities for human life.

As a notable example, vaccine development historically took years to decades, but in 2020 advances in computational immunology enabled the discovery of potential vaccine targets against COVID-19 within a few hours of the SARS-CoV-2 viral sequence being published, and just 6 months later several vaccine candidates were in clinical trials (Arnold, Nature Medicine 2020). Potential human health and disease applications include medical diagnostics, drug design, therapeutic decision-making and planning. Agricultural and (food) industry applications are also prominent, encompassing optimisation of crop and microorganism phenotypic traits.

Introduction 2

Courses:
 Q2: DSAIT4035 Algorithms for Sequence-based Bioinformatics
 Q3: DSAIT4040 Algorithms for Network-based Bioinformatics
 Q4: DSAIT4045 Machine Learning in Bioinformatics

DSAIT4035	Algorithms for Sequence-based Bioinformatics	5
Responsible Instructor	Dr. T.E.P.M.F. Abeel	
Responsible Instructor	Dr. J.A. Baaijens	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	After having followed this course, the student has a good understanding of algorithms and data structures in genomics used for DNA sequence analysis. The student is able to implement algorithms in python, and can translate methods described in scientific literature into a working implementation. Bioinformatics analyses in genomics aim to compare large sets of genomes in order to understand and explain differences in traits of an organism. Contemporary methods are powered by fundamental algorithms and data structures, which are efficient and scale to large data sets. A thorough understanding of these algorithms and data structures is necessary for advanced users and developers in this area. Bioinformatics application domains that are covered include: variant calling, motif detection, genome assembly, sequence alignment, DNA sequencing.	
Study Goals	At the end of this course, the student is able to: 1. Explain multiple sequencing technologies and their data properties. 2. Analyse core data structures and foundational computer science algorithms in sequencing-based applications of bioinformatics. 3. Implement conceptual data structures and algorithms with applications in genome sequence analysis. 4. Implement published algorithms in a real-life setting. 5. Benchmark a published sequencing-based algorithm. 6. Produce scientifically sound written and oral communication of a benchmark study that compares the students implementation against a literature implementation.	
Education Method	The course is offered as a mix of lectures, exercises and a project	
Literature and Study Materials	The course follows the course material from https://www.bioinformaticsalgorithms.org/	
Assessment	The final grade of the course consists of the following components: - Individual Assignments: programming exercises on foundational algorithms and data structures in bioinformatics sequence analysis (weighting 50%) - Individual Report: reflection essay on group project product (weighting 10%) - Group Report: report on the benchmarking project including the student implementation and a literature derived implementation (weighting 40%) Final grade calculation = $0.5 * \text{Individual Assignments} + 0.1 * \text{Individual Report} + 0.4 * \text{Group Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Assignments: Repair opportunity - Individual Report: Repair opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4040	Algorithms for Network-based Bioinformatics	5
Responsible Instructor	Prof.dr.ir. M.J.T. Reinders	
Responsible Instructor	J.M. Weber	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course will provide a brief overview on molecular biology, the advent of high throughput measurement techniques and large databases of biological knowledge. With this given background, the course will dive into the importance of various network approaches to model biological knowledge. We will cover topics such as complex network models to characterise and analyse biological systems, approaches to infer network structures from given biological measurements, strategies of network enhancement through network integration, predictions based on network structures, and graph generation (molecular design). Specifically, the course contains the following topics: Background on molecular data: systems biology, data-driven approach, high throughput measurements, sequencing, mass-spectrometry, immunoprecipitation, single cell measurements, molecular properties, molecular descriptors, databases. Molecular interactions: transcription factors, protein complexes, metabolic reactions, gene-protein-metabolite interactions, interaction models. Frameworks to work with complex systems: graphs with nodes/edges, directionality, layers, heterogeneous graphs, topological graph metrics, graph characteristics (erdos-renyi/scale-free/hierarchical networks), hypergraphs, topological hypergraph metrics, simplicial complexes, simplicial complex metrics Network inference and simulation: rate-law modelling of RNA regulation model, dynamic system/steady state, reduce, association networks, linear networks, design matrix, bayesian networks, boolean networks, regularisation, evaluation Network enhancement: heterogenous networks, reliability of networks, bias/variance, evaluation strategies, early/intermediate/late integration, machine learning approach, decision trees, classifier combining, probabilistic/dissimilarity/kernel-based representations, network fusion, random walks. Network-based predictions on the node-level: guilt-by-association, maximise coherency, Markov random field, clustering based on graph properties, Markov clustering, spectral clustering, Louvain method, active modules, validation. Network-based predictions on the graph level: graph neural networks, graph convolution, node message passing, graph attention, molecular property prediction, structure-based function prediction. Graph generation for molecular design: generative AI for molecular design, graph variational autoencoders, graph diffusion models	
Study Goals	At the end of this course, you should be able to: Give examples of the basic elements of a living cell, their interactions, and ways how the interactions can be measured and why computational models are used to model the interactions. Explain the types and functional principles of different mathematical models used to represent and model network problems Examine how network models can be used to analysis, enhance, mine, and infer information to gain further insights into a specified biological problem Argue upon recent network-based computational biology literature paying attention to proposed methods, advantages, and disadvantages, potential modifications of the method and alternative biological applications Formulate your own written network-based research proposal with a biology-oriented research problem, a motivated and innovative algorithmic solution, and well-reasoned evaluation strategies; in context of the materials presented throughout the course	
Education Method	The course consists of a mixture of lectures by the teachers and paper presentations by one or more of the students. Each paper presentation will be followed by an in-depth (plenary) discussion within the whole group. The students get iterative individual feedback on the research proposal they have to write through individual Q&A slots throughout the quarter.	
Literature and Study Materials	Lecture slides, papers	
Assessment	The final grade of the course consists of the following components: - Individual Report (weighting 80%) - Group-based Presentation on a scientific paper (weighting 20%) - Submission of paper discussion points (pass/fail) Final grade calculation = $0.8 * \text{Individual Report} + 0.2 * \text{Group-based Presentation scientific paper}$ A passing final grade for the course can only be earned when in both components at least a 5.0 is earned, and the weighted final grade is at least a 5.8, and the discussion point submission is a pass. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Report: Repair opportunity - Group Presentation scientific paper: Repair opportunity (individual or group) - Discussion points on paper: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4045	Machine Learning in Bioinformatics	5
Responsible Instructor	Dr. J.S. de Pinho Gonçalves	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Learning from patterns in molecular biology data plays an important role in diagnosing disease, discovering new targets for therapy, and more generally in answering biological questions that lead to an improved understanding of biological systems with relevance to human health, industry, biotechnology, and agriculture. This course focuses on methodology for the analysis of high-dimensional data in molecular biology, naturally addressing challenges that arise in the field such as learning from unlabelled data or from small numbers of samples. The methodology is introduced in the context of real-world applications with examples using real data. Covered topics will be (a selection of) the following: - Mixture models: (in)finite mixture models, EM algorithm, ECDF, bootstrapping. - Clustering: k-means/medoids, density-based clustering, hierarchical clustering, clustering evaluation, choosing number of clusters. - Statistical hypothesis testing: p-values, confidence intervals, multiple hypothesis testing, family-wise error rate, (local) false discovery rate. - Testing using high-throughput count data: multifactorial designs, (generalized) linear models, analysis of variance, shrinkage estimation. - Linear and non-linear dimensionality reduction: matrix decomposition (SVD), biplot representations, PCA, NMF, UMAP, autoencoders. - Multivariate methods for heterogenous data: embeddings, multidimensional scaling (MDS), robust (non-metric) MDS, correspondence analysis, canonical correlation. - Supervised learning: algorithms, performance metrics, curse of dimensionality, generalizability and model complexity, regularization, cross-validation, ML architectures for sequential data. - Interpretability and explainability. - Other selected advanced topics, including transfer learning self-supervised learning, semi-supervised learning, multimodal ML.	
Study Goals	At the end of this course, the student is able to: 1. Analyse different kinds of high-throughput bioinformatics data and their statistical properties. 2. Discuss published statistical methods and ML algorithms, and the data-driven bioinformatics problems they are designed to solve. 3. Formulate research questions or data-driven problems related to real-world challenges in the field of bioinformatics. 4. Design a scientifically well-reasoned data-driven solution to address problems / research questions in bioinformatics. 5. Evaluate the technical quality and scientific value of data-driven bioinformatics solutions. 6. Communicate on the results of a complete data-driven bioinformatics research project via a scientific report and presentation.	
Education Method	The course is run in a flipped classroom setting, with lectures and project labs: - Lectures: students take turns explaining the material from different book chapters or papers. All students read and prepare the material before every lecture, to be able to answer questions or small assignments (e.g. via a graded in-class quiz) and contribute to the discussion in the classroom. - Project/labs: students work in small groups and individually on a research project throughout the course. Students present on the status of the project (and receive formative feedback) halfway through the quarter, and deliver a written report+code at the end.	
Literature and Study Materials	- Holmes S. and Huber W. (2019) Modern Statistics for Modern Biology. Cambridge University Press. ISBN: 9781108705295 - Other selected book chapters and papers provided and updated with each edition of the course.	
Assessment	The final grade of the course consists of the following components: - Report grade: report about the bioinformatics project performed throughout the course (weight 60%) - Presentation grade: individual presentation on the content of a book chapter or scientific paper (weight 20%) - Assignments grade: small assignments/quizzes on the material of the course, the precise format of this assessment component will be communicated during the first lecture (weight 20%) The report grade itself consists of two parts, a group part (40% weight) and an individual part (60% weight). Final grade calculation = 0.6 * Report + 0.2 * Presentation + 0.2 Assignments A passing final grade for the course can only be earned when at least a 5.0 is earned per component, for all components, and the weighted final grade is at least 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Report: repair opportunity available only for the individual part. - Presentation: repair opportunity available. - Assignments: no repair.	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). Dr. T.E.P.M.F. Abeel	

Data and Information Management 2024	
Introduction 1	Data and information management is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, structured, and unstructured data is becoming more and more challenging. In the context of AI, solid data and information management is one of the core prerequisites of most AI and Machine Learning applications; it is often accepted that up to 80% of the efforts to deploy and effective AI system typically are spent on managing and preparing the required data. This theme introduces the principles and techniques of Web Science and Engineering, Information Retrieval, and Web-scale Data Management. The Web is now the primary source of data and information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, stored, analysed, and exploited so that the needs and desires of users can be satisfied in various contexts. The three courses provide the fundamental knowledge related to each topic, including the concepts, methods, and techniques, and highlight the research and practical challenges in the design, development, and usage of data and information management systems in contexts. Through the lectures, assignments, and the group project, students will gain an understanding of scientific and practical challenges in data and information management, and learn the skills to design, develop, and evaluate their own systems, from both a technological and human point of view.
Introduction 2	Courses: Q2: CS4525 Web-Scale Data Management Q3: DSAIT4050 Information Retrieval Q4: DSAIT4055 Web Science and Engineering

CS4525	Web-Scale Data Management	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. A. Katsifodimos	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: - BSc in Computer Science (or equivalent degree), or - Basic Knowledge of Data Management and (relational) Databases. - Programming Skills Recommended: Knowledge of Operating Systems, Networking Systems, Distributed Systems, Implementation of relational databases.	
Course Contents	This course addresses the challenges of Data Management at Web-scale. Especially, it covers the need for large-scale distributed data storage systems. The lecture therefore introduces step-by-step increasingly complex distributed storage systems, leading up to modern implementations of different NoSQL data storage systems. The challenges arising from such systems are presented and discussed, especially focusing on the CAP theorem and the resulting trade-offs with respect to data models, transactional power, query expressivity, and replication consistency. These discussions lead to different variants of NoSQL database systems, like Key-Value Stores, Document Stores, Wide-Columnar stores, and Graph Databases. The advantages, disadvantages, and general properties of these systems are discussed in more detail. The course focuses also on database transactions, and the implications those have in modern, Web-scale application development and deployment, i.e., there is special focus on distributed transactions and consistency guarantees of different data management systems and methods.	
Study Goals	At the end of this course, the student is able to: 1. Explain the trade-off between data consistency and scalability of data processing techniques. 2. Design a scalable data management architecture 3. Analyse a use case using high-level data management and use-case requirements 4. Evaluate a self-designed scalable data management architecture with experiments 5. Apply scalable data management techniques and methods in practical scenarios	
Education Method	The course comprises weekly lectures that introduce core data management concepts that are then exemplified via the presentation of real-world web-scale data management systems. From week 2, a project is announced, that the students can complete in teams of 3-5 persons. At the beginning of each lecture, the lecturer and the students discuss issues the projects, alongside summative feedback on progress as well as specific suggestions about the project. At the end of the class, the students present their project results, and the instructor also executes the project in a computing cluster (or the Cloud) infrastructure.	
Literature and Study Materials	Primarily course slides, and research papers. Recommended book: "Designing Data Intensive Applications", by Martin Klepmann.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 80%) - Group Assignments (weighting 20%) Final grade calculation = $0.8 * \text{Written Exam} + 0.2 * \text{Group Assignments}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4050	Information Retrieval	5
Responsible Instructor	Dr. A. Anand	
Responsible Instructor	Dr. M.S. Pera	
Instructor	J. Yang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: - BSc in Computer Science (or equivalent degree), or - Programming Skills and - Linear Algebra Recommended: - Natural Language Processing Course	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of IR-related solutions. Covered topics include: - Information Retrieval Models - Indexing techniques - Web Search - Evaluation of information retrieval systems - Recommender Systems	
Study Goals	At the end of this course, the student is able to: 1. Relate to basic concepts and terminology in IR 2. Distinguish different IR tasks that can be applied to solve problems in different information seeking scenarios 3. Create and execute a strategy for designing solutions to IR specific problems and challenges	
Education Method	Lectures, project-based learning	
Literature and Study Materials	Lectures, lecture notes, and supplementary reading material provided in class	
Assessment	The final grade of the course consists of the following components: - Individual Assignment: foundational IR tasks - text processing and matching (weighting 10%) - Individual Report (weighting 20%) - Group Report (weighting 70%) Final grade calculation = $0.1 * \text{Individual Assignments} + 0.2 * \text{Individual Report} + 0.7 * \text{Group Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Assignments: Repair opportunity - Individual Report: Repair opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4055	Web Science and Engineering	5
Responsible Instructor	J. Yang	
Responsible Instructor	Dr. C. Lofi	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: - BSc in Computer Science (or equivalent degree), or - Programming Skills Recommended: - Scientific Writing	
Course Contents	The main subject of the course is the Web and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	At the end of this course, the student is able to: 1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g., Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies	
Education Method	Lectures and supervised group-based project	
Literature and Study Materials	Research Papers and custom slide-deck	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 30%) - Group Report (weighting 70%) Final grade calculation = $0.3 * \text{Written Exam} + 0.7 * \text{Group Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Human-Centred AI 2024	
Introduction 1	AI offers a vast potential for addressing significant societal issues, including health, climate change, finance, and mobility, by utilizing an amount of data that was never before available. However, to utilize AI effectively, it is necessary to discover novel methods of integrating the analytical capabilities of computers with the power of human intelligence. AI solutions operating in the human world can benefit from understanding what humans do and want, especially if they directly interact with humans. Human-centred AI (HCAI) is a paradigm for designing, developing, and deploying AI-infused systems to benefit from and augment human intelligence and therefore place the human central in designing these solutions. For this, the theme teaches knowledge and skills about human behaviour and cognition, human-AI interaction, and the design and evaluation of human-AI solutions. We divide the education on HCAI into three courses where we start from a passive observational stance, where AI is designed or used to interpret the social human world and human cognitions, followed by a more interactive stance, and finally towards a proactive stance, where the AI actively intervenes in the human world to guide a human with specific intent, for example, to lead a healthier life. Students that complete this theme will be able to work on solutions such as developing acceptable interactive AI-infused systems that augment human intelligence, interpreting human behaviour and cognitions by automated approaches, and creating social agents, that can engage in face-to-face conversation with a human. They will be able to reason and argue about the social, cognitive, and technical challenges that accompany such solutions for users. For each course and throughout the theme, there are three categories of knowledge HCAI students must acquire. 1. Human and social factors for HCAI: This includes knowledge from disciplines such as sociology, psychology, and philosophy, on topics such as perception of social and affective phenomena, and human values and motivation that are relevant to developing interactive AI-infused systems. 2. HCAI practices, principles, and techniques: This includes, Human-AI interaction, collaboration, and synthesis, for example, human factors-informed AI models to represent conceptual human factors as computational abstractions, dialogue management and also algorithmic nudging. 3. Design, evaluation and engineering tools, techniques, and methodologies: This includes knowledge of tools, techniques, and methodologies necessary for designing and evaluating HCAI systems.
Introduction 2	Courses: Q2: DSAIT4060 Human-centred Machine Perception Q3: DSAIT4065 Conversational Agents Q4 DSAIT4070 Designing Human-Centred AI Systems

DSAIT4060	Human-centered Machine Perception	5
Responsible Instructor	Dr. H.S. Hung	
Responsible Instructor	Dr. B.J.W. Dudzik	
Instructor	Dr. C.A. Raman	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The ability to automatically perceive and interpret human behavior is crucial for adaptive intelligent technology (e.g., social robots or tutoring systems). Not only can it improve user understanding in applications, but it also holds the potential for novel scientific insights about human behavior in the real-world. This course covers relevant theories and principles for developing Human-centred Machine Perception systems that leverage sensor data (e.g., from cameras, microphones, or wearables) in individual, dyadic, and group settings. Topics covered include data collection, annotation, implementation, and evaluation of such systems, as well as relevant theories about human behavior and psychology (e.g., vocal expression, group dynamics, and human emotion).	
Study Goals	At the end of this course, the student is able to: 1. Explain behavioral science theories on human cognition, affect, and social interaction relevant to common applications of machine-human perception that were introduced in the course. 2. Explain the relation between behavioral science theories that were introduced in the course and the design of Human-centered Machine Perception systems. 3. Apply the advanced machine learning techniques that were introduced in the course for Human-centered Machine Perception in individual, dyadic, and group-level settings. 4. Critique all stages in the research and development pipeline of human-centered machine perception systems with respect to behavioral science and technical theories that were introduced in the course.	
Education Method	The course offers a series of lectures on theories, principles and methods, discussed and illustrated with examples. In class, student groups present summaries of assigned research articles on a selected topic in Human-machine Perception and potential limitations regarding data collection, annotation, training, and modeling. The lecturer gives formative feedback. A series of assignments focusing on advanced machine learning techniques for Human-centered Machine Perception tasks establish practical relevance.	
Literature and Study Materials	Burgoon, J. K., Magnenat-Thalmann, N., Pantic, M., & Vinciarelli, A. (Eds.). (2017). Social signal processing. Cambridge University Press. Picard, R. W. (2000). Affective computing. MIT press.	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 50%) - Individual Assignments: assignments applying advanced machine learning techniques in clearly scoped Human-centred Machine Perception tasks (weighting 50%) Final grade calculation = $0.5 * \text{Digital Exam} + 0.5 * \text{Individual Assignments}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4065	Conversational Agents	5
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Responsible Instructor	Z. Yue	
Instructor	S. Tan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Chatbots, embodied and conversational virtual agents, and social robots are becoming more and more popular. Many people are owning an Alexa, Cortana or Echo or are talking to their virtual assistant on their phone. Indeed, such technologies have the potential of making our lives easier and relieve people from the more repetitive tasks. For example, it is imaginable that such systems are being used for financial applications by helping customers with frequently asked questions but also to advise them on in the long term more impactful decisions such as their pension plans. Further applications can be imagined in the area of healthcare and education, some of which are already in existence today. In this course, attention will be given to different verbal and nonverbal behavioral characteristics of conversation and embodied interaction, such as, intonation, gaze and gestures that humans show when communicating with both other people and machines. This behavior is then related to different multimodal dialogue functions, including turn-taking, addressing others, and backchanneling, that give shape to the communication process. Topics on interaction with embodied agents and how to design memory models for conversational interaction are also discussed. Students apply this knowledge through the design, development, and evaluation of an embodied conversational agent application that uses a memory model to interact.	
Study Goals	At the end of this course, the student is able to: 1. Apply relevant linguistic, speech and psychological theory to conversational agent systems 2. Design the conversational memory of your agent focusing on a perception/generational model 3. Analyse human-human conversational data to better design ML models for adding generation or perception capabilities to conversational agents 4. Develop the conversational memory in your agent 5. Quantitatively evaluate a conversational agent in a user-experiment. 6. Analyse your results and the limitations of your implementation	
Education Method	There are 2 lectures scheduled per week. Students work in groups of 4 on a group project. There are 3 labs that focus on components needed for the project.	
Literature and Study Materials	"The conversational interface " by Michael McTear, Zoraida Callejas, David Griol. This book is freely available through the TU Delft library. Also, recent papers selected by instructors.	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 30%) - Group Report: Group report on the design, implementation, and evaluation of a conversational agent (weighting 70%) - Group Assignment 1: group lab exercise - memory (pass/fail) - Group Assignment 2: group lab exercise - perception (pass/fail) - Group Assignment 3: group lab exercise - evaluation (pass/fail) Final grade calculation = $0.3 * \text{Digital Exam} + 0.7 * \text{Group Report} + \text{Pass/Fail} * \text{Group Assignment 1} + \text{Pass/Fail} * \text{Group Assignment 2} + \text{Pass/Fail} * \text{Group Assignment 3}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Group Report: Repair opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4070	Designing Human-Centred AI Systems	5
Responsible Instructor	M.L. Tielman	
Responsible Instructor	Dr.ir. U.K. Gadiraju	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: Basic programming skills (e.g., CSE1500, CSE1100), or BSc Computer Science (or equivalent).	
Course Contents	Recommended: prior experience in algorithm design (e.g., CSE2310), Theory Probability and statistics (e.g. CSE1201), introduction level machine learning (e.g. CSE2510, CSE2530) In today's world, interactive AI-infused systems like movie and music recommendation systems, as well as health or financial digital advice systems, provide algorithmic support to individuals in achieving their short- and long-term goals. Ideally, these systems aim to align with people's values, motivations, habits, and social context, offering augmentation of capabilities by providing algorithmic solutions to tasks that may be mentally demanding or require expertise beyond the individual's reach. As automation levels rise, ensuring meaningful control, trustworthiness, and understandability of these systems becomes crucial for user acceptance. In this course, students learn to design and evaluate these systems from a human-centred AI perspective. The course incorporates psychological theories and concepts that inform algorithmic solutions to nudge individuals towards making good decisions or enhancing their ability to modify their behaviour without coercion or deception. By understanding human behaviour and cognitive processes, students will learn how to design AI systems that support users in a beneficial, transparent, and respectful way of their autonomy. The course's main topics: - Human motivation, values, cognition, behaviour, learning, change, decision-making, and persuasion - HCAI design methods - Technology acceptance of interactive AI-infused systems - Design guidelines for Human-AI interaction - Algorithmic nudges and boosts	
Study Goals	At the end of this course, the student is able to: 1. Apply psychological theories, principles, and concepts on human motivation, values, cognition, behaviour, learning, change, decision-making, and persuasion in the design of AI-infused systems that interact with humans. 2. Apply Human-Centred AI theories, principles, methods and concepts of acceptable AI, responsible AI, trustworthy AI, persuasive technology, user mental models, and AI design metaphors in the design of interactive AI-infused systems. 3. Construct an implementation of a responsible and explainable algorithmic nudge, boost, or persuasion for a specific real-world application domain and objective. 4. Design an interactive AI-infused system that is usable and beneficial to people using Human-Centred Design methods and techniques. 5. Evaluate an interactive AI-infused system from a human-centred perspective.	
Education Method	The course offers a series of lectures (combination of pre-recorded video and live lecture sessions) on theories, principles and methods, discussed and illustrated with examples. In the class, student groups present their progress on the assignment and receive formative feedback from the lecturer. Outside the lectures, students work in small groups on the assignment and take digital self-quizzes to the material presented in the lecture.	
Literature and Study Materials	Selected papers & chapters will be used for the different lecture topics. All reading material will be made available on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Group Presentation: presentation of the design and evaluation of interactive AI-infused system including demonstration of computer simulation of algorithmic nudge, boost or persuasion for this system (weighting 30%) - Individual Presentation: individual Q&A session about design, evaluation of interactive AI-infused system and algorithmic nudge or boost for this system (weighting 20%) - Digital Exam (weighting 50%) Final grade calculation = $0.3 * \text{Digital Presentation} + 0.2 * \text{Individual Presentation} + 0.5 * \text{Digital Exam}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Presentation: Repair opportunity - Individual Presentation: Repair opportunity - Digital Exam: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr.ir. W.P. Brinkman	

Interactive Visual Analysis 2024	
Introduction 1	Interactive data science concerns data science and AI methods relying on a human in the loop. While automatic data analysis is desirable from a (cost) efficiency point of view, such pipelines have limited capacity when it comes to gaining new knowledge from data or forming new hypotheses based on a given dataset. With rapid advances in AI targeting analysis tasks, AI-based decision making relies on trust in and fairness of the underlying algorithms and their output. Such trust can be built with making the inner workings of AI models visible and letting analysts interact with them. Even with trustworthy AI, critical decisions will still often need to be made by humans (see also the EU AI Act). Interactive and visual data science is concerned with providing the right tools for human-in-the loop data analytics. This theme focuses on interactive aspects of data science with a human-in-the loop. Specifically, we will discuss principles from data visualization, visual analytics, human computer interaction, perception, and effective data communication and how they are applied to interactive data exploration in the context of data science. Based on these principles, the theme explores how visual data representations are tailored to different tasks and audiences, like medical practitioners. Furthermore, specific attention is directed towards the design and implementation of interactive data visualization pipelines that realize the discussed principles within a concrete application context. We divide the education on interactive data science in three courses. In the Data Visualization course, We start with a focus on visual analysis of heterogeneous with a human in the loop. This includes design study methodology, building on the principles of effective visual communication using human visual perception principles, effective visual representations for different data types and analysis tasks, meaningful interaction patterns for explorative data analysis and more. In the second course, 3D Visualization, the focus will be on efficient algorithms for real-time visualization of complex spatial data, such as medical scans or scientific simulations. Finally, the third course on High-Performance Analysis Systems builds on top of the previous courses to combine high performance algorithms with effective human computer interfaces to build complete visual data analysis systems for real-world problems.
Introduction 2	Courses: Q2: DSAIT4075 Data Visualization Q3: DSAIT4080 3D Visualization Q4: DSAIT4085 High-Performance Analysis Systems

DSAIT4075	Data Visualization	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Recommended: Programming skills in javascript/html/css. The D3 library will be used. It is advisable to work through a tutorial before the course. E.g., https://github.com/sgratzl/d3tutorial	
Course Contents	Data visualization is the visual representation of data by computer generated images. Here we focus on human principle for visual analysis of various data collections such as databases. The goal is to improve insight, understanding and/or communication of data. Data visualizations use a combination of methods from a very diverse variety of disciplines: perception, computer graphics, human computer interaction, machine learning, numerical analysis, optimization, etc. Topics covered: models of the visualization process; colour models and use of colour; information visualization; representation and processing of data; interactive visual data analysis; truthful data visualization.	
Study Goals	At the end of this course, the student is able to: 1. Discuss a large range of visualization techniques. 2. Discuss principles of human perception and cognition in visual data analysis. 3. Design visual analysis systems for practical data analysis problems. - Analyse a data analysis problem, identify its suitability for visual analysis and abstract requirements. - Implement a suitable visualization solution. - Assess the quality and performance of the implemented solution. 4. Assess the truthfulness of data representation.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of data visualization, common problems and their solutions. Various algorithms and their properties will be discussed. You will work on a project throughout the quarter in groups during the lab sessions (2 hours each) and at home. You will receive formative feedback on your approach during the lab sessions.	
Literature and Study Materials	"Visualization Analysis and Design", Author: Tamara Munzner, CRC Press	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 40%) - Group Assignment: data visualization project (weighting 60%) Final grade calculation = 0.4 * Written Exam + 0.6 * Group Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade (rounded to .5) is at least a 6. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity in form of a new individual project during Q3 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4080	3D Visualization	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	T. Höllt	
Instructor	Dr. P. Kellnhofer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: Computer Graphics (CSE2215), Programming skills in C++, Open GL (a crash course in OpenGL will be provided)	
Course Contents	Theory and practice of spatial data visualization such as medical scans or simulations. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measurement; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for 3D; advanced applications and specific requirements for different fields, such as medical.	
Study Goals	At the end of this course, the student is able to: 1. Explain 3D visualization algorithms and their applications for example in medicine. 2. Discuss the advantages and disadvantages of 3D-oriented visualization. 3. Discuss implications with regards to ethical issues, trust, etc. with a particular focus on medical visualization systems. 4. Build a 3D visualization system for a given problem.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of data visualization, common problems and their solutions. Various algorithms and their properties will be discussed. During the lab sessions (2 hours each), you will work in groups on specific data visualization projects and you will receive formative feedback on your approach.	
Literature and Study Materials	"Visual Computing for Medicine", Second Edition: Theory, Algorithms, and Applications, Authors: Bernhard Preim and Charl P. Botha	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 40%) - Group Assignments: implementation assignments (weighting 60%) Final grade calculation = $0.4 * \text{Written Exam} + 0.6 * \text{Group Assignments}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade (rounded to .5) is at least a 6. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4085	High-Performance Analysis Systems	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	Dr. M. Skrodzki	
Instructor	Dr.ir. U.K. Gadiraju	
Instructor	Prof.dr.ir. M.B. van Gijzen	
Instructor	Dr. D. Palagin	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	High-Performance Analysis Systems cover systems for user interaction with data that puts a special emphasis on performant implementations. The data sets can be results of numerical simulations or measurements (scientific visualization), or other data collections such as databases (information visualization). The goal is to improve insight into, understanding of, and/or communication about the data. Therefore, three parts are taken into account: (1) Computer-Human Interaction: How to design effective and efficient interfaces? (2) High-Performance Computing: How to program algorithms that can process and visualize (large) data sets fast? (3) Analytics and Applications: How to derive conclusions from data based on the algorithms implemented?	
Study Goals	At the end of this course, the student is able to: 1. Explain basic concepts of high-performance computing. 2. Understand the concepts and theories of human-computer interaction relevant to data visualization. 3. Compare human-centred computing methods involving users in various stages of the design and development of visualization techniques. 4. Create high-performance computing solutions with a data-centric focus. 5. Create a high-performance computing solution to an interactive explorative visualization problem.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of data visualization, common problems and their solutions. Various algorithms and their properties as well as efficient visualizations will be discussed. Furthermore, the human within the loop is taken into account. During the lab sessions (2 hours each), you will work in groups on building the final project and you will receive formative feedback on your approach.	
Literature and Study Materials	"A Practical Approach to High-Performance Computing", Springer, authors: Sergei Kurgalin, Sergei Borzunov. "Visualization Analysis and Design", A K Peters Visualization Series, CRC Press, author: Tamara Munzner.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 25%) - Group Assignment 1: visualization project - visualization algorithm (weighting 25%) - Group Assignment 2: visualization project - computer-human interaction (weighting 25%) - Group Assignment 3: visualization project - high-performance computation (weighting 25%) Final grade calculation = 0.25 * Written Exam + 0.25 * Group Assignment 1 + 0.25 * Group Assignment 2 + 0.25 * Group Assignment 3 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade (rounded to .5) is at least a 6. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Language and Speech Technology 2024	
Introduction 1	Language and speech technology concerns the processing, analysis, and generation of human language data with computers. Human language has a direct role, either in text or speech, as an important medium enabling interaction between humans and computers, robots, and machines; language data, is also a major type of data AI interprets. Recent advances in Natural Language Processing and Speech Technology have brought language and speech technologies to the forefront of AI. These technologies are at the core of a wide range of AI applications, from digital services such as search engines, voice assistants, to domain-specific applications in health, finance, design, education and many others. This theme introduces the principles and techniques of Natural Language Processing (NLP), Speech Technology, and Societal Applications of NLP in three separate courses. The three courses provide the fundamental knowledge related to each topic, highlight the scientific questions, and focus on the practical challenges in the design and development of the technologies but also those arising from their use in contexts, including robustness as well as the social and ethical issues. Through the lectures, assignments, and the group project, students will gain an understanding of classic and modern techniques for language and speech processing, understanding, generation and how they are combined to develop (spoken) dialog and deliberation-support systems, and learn the skills to design, develop, and evaluate their own language, speech, and dialog applications, from a technological, human, and ethical/societal point of view.
Introduction 2	Courses: Q2: DSAIT4090 Natural Language Processing Q3: DSAIT4095 Inclusive Speech Technology Q4: DSAIT4100 Natural Language Processing for Society

DSAIT4090	Natural Language Processing	5
Responsible Instructor	J. Yang	
Responsible Instructor	Dr. A. Anand	
Instructor	Dr. C. Lofi	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Natural Language Processing (NLP) concerns the interaction between computers and human language, in particular how to build systems to process and analyze large amounts of human language data. It has become an important technology in many parts of computer science, facilitating key applications of large-scale text understanding, automatic question answering, knowledge base construction, information retrieval, text or speech-driven dialogue agents, translation, text summarization, etc. This course aims to introduce the principles and techniques for Natural Language Processing. Through the lectures, assignments, and the group project, students will gain an understanding of classic and modern techniques for language understanding and generation and learn the skills to design, develop, and evaluate their own Natural Language Processing systems and applications.	
Study Goals	At the end of this course, the student is able to: 1. Recognize typical NLP tasks and components of an NLP system. 2. Describe typical lexical, syntactic, and semantic analysis techniques. 3. Compare token embedding techniques, common neural architectures (e.g., attention), and their implementation in neural language models. 4. Compare deep contextual embeddings techniques (e.g., GPT and BERT). 5. Describe information extraction techniques and its application to NLP tasks. 6. Analyse the implications of the design choices in language data processing and machine/deep learning pipelines in terms of transparency, accountability, and algorithmic fairness and harms. 7. Create a well-reasoned NLP system, possibly using neural methods with semantic enrichment and user adaptation functionalities. 8. Evaluate the relevance and correctness of the design choices.	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments.	
Literature and Study Materials	- Dan Jurafsky and James H. Martin. 2014. Speech and language processing. Pearson. - Yoav Goldberg. 2015. A Primer on Neural Network Models for Natural Language Processing. - Zteven Bird, Ewan Klein, and Edward Loper. 2009. Natural language processing with Python: analyzing text with the natural language toolkit. " O'Reilly Media, Inc.".	
Assessment	The final grade of the course consists of the following components: - Individual Report: individual literature survey and assignments (weighting 20%) - Group Report: group project, written report and interview (weighting 80%) Final grade calculation = 0.2 * Individual Report + 0.8 * Group Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Report: Repair opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4095	Inclusive Speech Technology	5
Responsible Instructor	Dr. O.E. Scharenborg	
Responsible Instructor	Z. Yue	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: Deep learning, Language and Speech Technology; Signal processing	
Course Contents	In Inclusive Speech Technology, students learn fundamental concepts, techniques, and applications in speech technology, with a strong focus on fairness and inclusivity, i.e., developing speech technology that can be used by everyone, irrespective of how one speaks. The course teaches students the fundamentals about human speech production and processing, and automatic speech processing and production techniques with a focus on statistical methods. The focus of the course is on developing skills for developing speech-based applications. Knowledge will be obtained through interactive lectures; while research skills will be developed in weekly labs in which concrete speech technology applications will be build.	
Study Goals	At the end of this course, the student is able to: 1. Explain speech production by humans and computers (text-to-speech synthesis), and how computers recognise speech with statistical methods. 2. Evaluate speech synthesis outcomes and speech recognition results. 3. Describe different methods used to produce and recognise speech with computers. 4. Design data collections for speech technology research and development. 5. Mitigate the implications of design choices in speech data processing and machine learning pipelines, transparency, accountability, algorithmic fairness and harms.	
Education Method	Weekly lectures and weekly labs. Over the course of the weeks, students will focus on a different aspect of the speech technology development process, from creating and processing data to training and evaluating the system.	
Literature and Study Materials	- Several chapters from Dan Jurafsky and James H. Martin. Speech and Language Processing (2nd edition) - Scientific articles - Videos	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 50%) - Individual Report: individual report on the term project describing the design, evaluation and discussion of an inclusive speech processing system (weighting 50%) Final grade calculation = $0.5 * \text{Digital Exam} + 0.5 * \text{Individual Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Individual Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4100	Natural Language Processing for Society	5
Responsible Instructor	L. Cavalcante Siebert	
Responsible Instructor	Dr. P.K. Murukannaiah	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Most societal applications involve multiple stakeholders. An important characteristic of such applications is the plurality of opinions on all aspects of the application from requirements to the desired outcomes. Language is the primary means in which the stakeholders express their opinions. Thus, language technology has an important role in facilitating constructive dialog and deliberation in such applications. NLP for Society course teaches concepts and methods necessary for building dialog and deliberation-support systems that can facilitate societal exchange. On the one hand, the course will focus developing NLP techniques for recognizing the core elements of a deliberation, including arguments, stance, values, perspectives, novel viewpoints, and disagreement. On the other hand, the course will focus on critical evaluation of dialog and deliberation-support systems from the perspective of deliberative ideas, including fairness, inclusion, and respect.	
Study Goals	At the end of this course, the student is able to: 1. Explain the key components of a dialog and deliberation-support system that facilitate societal discussions. 2. Explain the key human factors (such as values, stance, sentiment, and perspectives) that underlie societal dialog and deliberation. 3. Create data-driven techniques to ground human factors (such as values, stance, sentiment, and perspectives) in textual and spoken language. 4. Evaluate dialog and deliberation-support systems from a human-centred perspective. 5. Analyse ethical challenges such as fairness, accountability, and transparency in dialog and deliberation systems	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments.	
Literature and Study Materials	- Dan Jurafsky and James H. Martin. 2014. Speech and language processing. Pearson. - André Bächtiger and John Parkinson. 2019. Mapping and Measuring Deliberation: Towards a New Deliberative Quality. Oxford University Press.	
Assessment	The final grade of the course consists of the following components: - Individual Report: individual report on the term project describing the design, development, evaluation, and analysis of ethical impact of an NLP technique for dialog or deliberation support. (weighting 60%) - Individual Presentation: individual presentation, including Q&A, of the term project and individual assignment (weighting 10%) - Individual Assignment: individual reflections (written assignment) of existing works on Societal Applications of NLP (weighting 30%) Final grade calculation = $0.6 * \text{Individual Report} + 0.1 * \text{Individual Presentation} + 0.3 * \text{Individual Assignment}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Report: Repair opportunity - Individual Presentation: Resit opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Optimisation & Reasoning 2024	
Introduction 1	Many of the important problems in society we aim to address with support of artificial intelligence are combinatorial (NP-hard) optimisation problems. For example, health treatment planning, optimizing logistic operations, automated production, efficient planning for the energy transition, and scheduling of train operations are immensely important since this may directly result in lower costs, increased productivity, and/or higher quality. Although such problems may seem very different, the same approaches may be applied. In this theme, students learn how to formally specify deterministic problems in an unambiguous and rigorous way, apply advanced algorithms to solve these problems, and reason about the trustworthiness of the solutions.
Introduction 2	Courses: Q2: CS4530 Modelling and Problem Solving Q3: CS4535 Constraint Solving Q4: CS4205 Evolutionary Algorithms

CS4205	Evolutionary Algorithms	5
Responsible Instructor	Prof.dr. P.A.N. Bosman	
Responsible Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks.	
	This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, (GPU) parallelization, and real-world applications.	
Study Goals	At the end of this course, the student is able to: 1. Explain the key concepts underlying the main streams in Evolutionary Algorithm (EA) research, with in particular genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2. Explain key ingredients underlying the rationale of when these algorithms work and when they do not work. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3. Explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4. Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5. Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	8-10 lectures 4 smaller (individual) projects 1 larger group project	
Literature and Study Materials	Various scientific papers, course-specific texts, and slides	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment (weighting 30%) - Individual Assignments: weekly assignments (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.3 * Group Assignments + 0.1 * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4530	Modelling and Problem Solving	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. E. Demirovi	
Instructor	Ir. S.E. Verwer	
Instructor	Y. Murakami	
Instructor	S. Dumanı	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: bachelor course on Algorithm Design, programming Recommended: bachelor course on reasoning, complexity theory	
Course Contents	The course covers major (combinatorial) solving approaches, namely constraint programming, integer programming, SAT, decision diagrams, and local search. The course focuses on modelling practical problems, expressing them in the survey paradigms; algorithms for solving such problems are covered in the follow-up "Constraint Solving" course. Lastly, the course focuses on conducting proper empirical evaluations.	
Study Goals	At the end of this course, the student is able to: 1. Comprehensively explain main modelling paradigms such as constraint programming, mixed-integer programming, SAT and decision diagrams. 2. Motivate a choice for an appropriate modelling paradigm. 3. Model real-world problems in a suitable modelling paradigm. 4. Conduct proper empirical evaluation of their models and related algorithms.	
Education Method	Lectures and labs	
Literature and Study Materials	* Russell and Norvig, Artificial Intelligence: A Modern Approach, 4th (Global) Edition * provided study material	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 60%) - Group Assignment 1: lab assignment (weighting 10%) - Group Assignment 2: lab assignment (weighting 10%) - Group Assignment 3: lab assignment (weighting 10%) - Group Assignment 4: lab assignment (weighting 10%) Final grade calculation = $0.6 * \text{Digital Exam} + 0.1 * \text{Group Assignment 1} + 0.1 * \text{Group Assignment 2} + 0.1 * \text{Group Assignment 3} + 0.1 * \text{Group Assignment 4}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity - Group Assignment 4: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Special Information	For personal administrative matters only: mps-cs-ewi@tudelft.nl For general questions: please use answers.ewi.tudelft.nl	
Tags	Artificial intelligence Group work Modelling	
Co-Instructor	Prof.dr. M.M. de Weerd	

CS4535	Constraint Solving	5
Responsible Instructor	Dr. N. Yorke-Smith	
Responsible Instructor	Dr. E. Demirovi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Covers state-of-the-art algorithms used in practice to solve combinatorial optimisation problems, focusing on (Max)SAT and (lazy clause generation) constraint programming. The course offers an in-depth view of the algorithms including advanced techniques (e.g., propagator design, conflict analysis, decomposition approaches), alongside techniques that can certify the correctness of the output of the algorithms beyond reasonable doubt. The assignments include a notable programming component with Rust as the programming language. After the course, students will have a principled understanding of the algorithms and will be able to use that knowledge to design their own tailored approaches to solve combinatorial optimisation problems.	
Study Goals	At the end of this course, the student is able to: 1. Explain how combinatorial optimisation paradigms based on (Max)SAT, (lazy clause generation) constraint programming, and to some extent mixed-integer linear programming work. 2. Understand the main mechanisms behind ensuring the correctness of the output of algorithms (certificates of unsatisfiability / optimality). 3. Analyse the inner workings of the algorithms to determine the reason for the effectiveness of one paradigm over the other on a given problem. 4. Design and implement algorithms to solve combinatorial optimisation problems.	
Education Method	Lectures, practical assignments, labs	
Literature and Study Materials	Lecture notes, lecture videos, academic papers, package documentation, various textbooks.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment 1: lab assignment (weighting 10%) - Group Assignment 2: lab assignment (weighting 10%) - Group Assignment 3: lab assignment (weighting 10%) - Group Assignment 4: lab assignment (weighting 10%) Final grade calculation = $0.6 * \text{Written Exam} + 0.1 * \text{Group Assignment 1} + 0.1 * \text{Group Assignment 2} + 0.1 * \text{Group Assignment 3} + 0.1 * \text{Group Assignment 4}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity - Group Assignment 4: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Probabilistic Decision Making 2024	
Introduction 1	Decision making is a key skill of an intelligent agent: how should it choose its actions to accomplish its task as well as possible? A complicating factor is that many, if not all, real-world problems suffer from various forms of uncertainty, typically captured using probabilistic models. As such, in order to make effective decisions, intelligent agents need to learn and reason about the potential outcome and effects of their actions, as well as how to interpret the observations they receive. In the Probabilistic Decision Making theme, the so-called decision-theoretic approach to these challenges is considered, which means that we combine probabilistic (and causal) reasoning and utility theory to derive what actions a rational agent should take. The theme covers algorithmic techniques like model-based planning under uncertainty and reinforcement learning. Specific topics that we will treat include: - Representing uncertainty, probabilistic modelling - Reasoning about uncertainty with probabilistic inference - Reasoning about effects of actions and interventions: causal models and inference - Specifying complex tasks and dealing with multiple objectives - Planning under uncertainty - Reinforcement learning (including Bayesian RL and Deep RL) - Verifying safety under uncertainty
Introduction 2	Courses: Q2: DSAIT4105 Probabilistic Models and Inference Q3: DSAIT4110 Sequential Decision Making Q4: DSAIT4115 Deep Reinforcement Learning

DSAIT4105	Probabilistic Models and Inference	5
Responsible Instructor	Dr. F.A. Oliehoek	
Responsible Instructor	S. Dumani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course will focus on advanced probabilistic inference and modelling techniques, complementing the core 'Probabilistic AI & reasoning' course, and causality. On the inference side, the topics include: representation of uncertainty (Bayesian networks, undirected graphical models, probabilistic programs); exact inference algorithms for Bayesian networks; approximate inference algorithms (importance sampling, metropolis-hastings, particle filtering, variational inference); implementation strategies for probabilistic programs; learning for probabilistic inference; discrete probabilistic programming; types of probabilistic queries; algorithms for causal inference. On the modelling side, the course will include a spectrum of common probabilistic models that are often used in practice.	
Study Goals	At the end of this course, the student is able to: 1. Comprehensively explain key concepts of advanced probabilistic inference and causality. 2. Comparatively analyse strengths and weaknesses of advanced inference procedures. 3. Express real-world problems as probabilistic models. 4. Relate real-world problems to suitable techniques. 5. Evaluate self-implemented probabilistic inference techniques.	
Education Method	Lectures and lab assignments	
Literature and Study Materials	Materials based on the following books will be provided: - Russell and Norvig: Artificial Intelligence: A Modern Approach, Global edition. - David Barber: Bayesian reasoning and machine learning, Cambridge University press - van de Meent et al: Introduction to probabilistic programming	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 70%) - Group Assignment 1: assignment applying modelling methods (weighting 15%) - Group Assignment 2: assignment empirically evaluating methods (weighting 15%) Final grade calculation = $0.7 * \text{Written Exam} + 0.15 * \text{Group Assignment 1} + 0.15 * \text{Group Assignment 2}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4110	Sequential Decision Making	5
Responsible Instructor	Prof.dr. M.T.J. Spaan	
Instructor	L. Cavalcante Siebert	
Instructor	Dr. A. Lukina	
Instructor	I.M. Olkhovskaia	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Sequential Decision Making under uncertainty is an important skill of an intelligent agent, as planning ahead will allow it to achieve its goals better. This course will deepen the students' knowledge of AI decision making by focusing on the following core aspects. First, we discuss how the goals of an agent can be represented properly. Next, we delve into the exploration-exploitation trade-off in multi-armed bandits, a crucial concept for making decisions under uncertainty. Subsequently, we cover search and/or learning algorithms that use the Markov Decision Process (MDP) framework to compute (sub)optimal decisions in sequential settings. Extensions of the basic MDP framework that are relevant for real-world problems such as partial information, multiple objectives and acting in multiagent systems are also covered. Next, we discuss techniques to improve the safety and robustness of the decisions of agents. Finally, the gained theoretical knowledge is applied in 3 lab assignments, where students need to design and program an algorithmic solution for decision-making problems.	
Study Goals	At the end of this course, the student is able to: 1. Create suitable representations for specific tasks, using defined goals of an intelligent agent. 2. Apply appropriate planning-under-uncertainty or reinforcement-learning algorithms to sequential decision-making problems. 3. Apply model-based algorithms to address issues in sequential decision making, such as partial observability, multiple objectives, and presence of multiple agents. 4. Discuss issues regarding safety and robustness in sequential decision making and algorithmic techniques for verifying and safely learning decisions. 5. Apply existing sequential decision-making algorithms to real-world inspired problem domains.	
Education Method	Lectures, lab work (programming assignments)	
Literature and Study Materials	The course is based on several review papers.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 70%) - Individual Assignment 1: Lab assignment where students apply algorithms for (contextual) bandits (weighting 10%) - Individual Assignment 2: Lab assignment where students have to implement and evaluate a basic reinforcement learning algorithm (weighting 10%) - Individual Assignment 3: Lab assignment where students have to model the verification of a decision-making problem and use an existing verification tool (weighting 10%) Final grade calculation = $0.7 * \text{Written Exam} + 0.1 * \text{Individual Assignment 1} + 0.1 * \text{Individual Assignment 2} + 0.1 * \text{Individual Assignment 3}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Assignment 1: Repair opportunity - Individual Assignment 2: Repair opportunity - Individual Assignment 3: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. F.A. Oliehoek	

DSAIT4115	Deep Reinforcement Learning	5
Responsible Instructor	Dr. J.W. Böhrer	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course will cover the breadth of modern model-free RL methods, discuss their limitations and introduce a variety of current research topics. In particular, we expect to cover the following: - deep learning methodology and architectures - stabilization of approximated value estimation - modern actor-critic methods - planning as inference - exploration with deep networks - offline reinforcement learning - deep multi-agent reinforcement learning - multi-task and meta learning	
Study Goals	At the end of this course, the student is able to: 1. List the strengths and limitations of modern deep Reinforcement Learning approaches. 2. Explain the underlying concepts of the discussed methods, and how they differ from each other. 3. Derive the objectives and constraints of selected algorithm. 4. Implement selected algorithms/architecture.	
Education Method	Lectures, homework, tutorials, lab sessions (discussion of lecture and homework)	
Literature and Study Materials	Lecture slides	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) - Individual Assignments: Students hand in homework bi-weekly and must get 75% of the total points to be eligible to the exam (pass/fail) Final grade calculation = 1 * Written Exam + Pass/Fail * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Prof.dr. M.T.J. Spaan	

Year

Organization

Education

2024/2025

Electrical Engineering, Mathematics and Computer Science

Master Data Science and Artificial Intelligence Technology

Software Engineering for DS and AI 2024	
Introduction 1	Moving from a prototype AI model to a production system is very challenging and requires engineering skills. This theme focuses teaching students to apply engineering methods and best practices for AI and Data-driven systems. It will cover architectural design principles, continuous integration (CI) and continuous delivery (CD) pipelines, validation and testing techniques for AI systems, design and assess non-functional properties, including privacy, energy consumption, response time, and robustness.
Introduction 2	Courses: Q2: CS4570 Machine Learning for Software Engineering Q3: CS4575 Sustainable Software Engineering Q4: CS4295 Release Engineering for Machine Learning Applications

CS4295	Release Engineering for Machine Learning Applications	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	While the course does not have formal pre-requisites, some skills are expected: - You are able to use basic Git commands - You have a basic understanding of networking concepts - You are not afraid of using a command line interface The course will provide introductory material for self-study that can serve as a refresher or provide pointers for students to fill knowledge gaps.	
Course Contents	The world of Software Engineering has been revolutionized in the last decade. Instead of releasing software updates yearly, companies can now release multiple times per week, sometimes even per day, to their customers. This allows quick reactions to market demands, software failures, and it is crucial for increasing the business value of software. These improvements have been mostly enabled by advances in release engineering and, in this course, we will learn about the techniques and technologies that build the foundation for modern release engineering. This course is addressing two main target audiences: 1) software engineers that would like to learn how to systematically release their applications and make them scalable in an orchestrated environment and 2) machine learning experts that would like to learn how to make their proof-of-concept applications production ready. Together, we will go on a journey that starts at continuous integration and then moves on to continuous delivery, continuous deployment, and continuous experimentation. We will discuss the theory and the current research on various related subjects like containerization, testing, or monitoring and will put the learned theory into practice. As a running example, we will build a pipeline for a machine learning application, which -compared to traditional release engineering- poses additional challenges, like data versioning or model deployment. *Important:* Make sure to register for a group in Brightspace AT THE LATEST during the first lecture. *Important:* The course introduces several state-of-the-art tools that are used in industry. While -in theory- all major operating systems are supported, experience has shown that Windows users will face challenges. To reduce the friction points, we recommend affected students to either dual-boot a Linux system or to make use of a Linux Live system via USB when working on the exercises and assignments.	
Study Goals	After following this course, you will be able to... 1) Explain the concepts of DevOps and MLOps 2) Automate the development and operation process ... Automate packaging, versioning, and releasing in modern build environments like GitLab or GitHub ... Release artifacts in public registries like the GitHub Maven Registry. ... Provision servers with Vagrant and Ansible ... Automate the model training pipeline using DVC 3) Use modern containerization technology to build a scalable micro-service architecture ... Use Docker to isolate a complex, web-based application in a container ... Use Kubernetes to orchestrate a non-trivial, distributed application ... Create an Istio service mesh in Kubernetes that enables custom routing strategies ... Use Prometheus and Grafana to monitor the state of a Kubernetes-based application ... Find and understand documentation of open-source tools 4) Deploy and maintain machine learning models in production ... Embed a typical ML model in an HTTP micro-service ... Apply version-control techniques to data and models in a machine learning project ... Implement quality-control techniques in a machine learning application ... Design a testing strategy that covers data, model, infrastructure, and monitoring aspects 5) Document the state and extensions of release engineering pipeline ... Find related work and inspiration for release engineering tasks ... Identify extension opportunities in existing release pipelines	
Education Method	- Follow interactive lectures and pre-recorded material for self-study - Execute a programming project in small teams - Transfer lecture contents into a project through practical assignments - Assess the work of yourself and others through rubric-supported peer-feedback - Actively participate in Q&A sessions - Document your release pipeline in a report - Present your project results and answer questions about the lecture contents.	
Literature and Study Materials	- Lecture slides and recordings - In-class exercises - Tool documentation - External articles - Research papers	
Assessment	--- Formative Assessment --- You can check your individual learning progress through weekly rubrics ... You will receive feedback from other groups as a neutral outsider ... You can check your own progress via self-assessment Joint Q&A sessions will discuss common points of the peer feedback and concrete questions --- Individual Knock-Out Elements --- The course enforces several criteria to protect the active members of a team. Students not reaching the following criteria will be removed from the course:	

Upholding a professional attitude when interacting with peers and staff.
Actively participating in the team process and contributing to all team deadlines and deliverables
Constructive participation in all peer reviews for assignments (one for another team and one for your own)
Meaningful and visible project contribution throughout each of the seven project weeks (W2-W4, W6-W9)
... Create 1+ mergeable PR with meaningful* contribution
... Approve 1+ PR that is not your own
... Failing to satisfy the criteria in two subsequent weeks will lead to a course fail
At the end of the course, you must meet the following criteria
... 7+ meaningful* PRs of you were merged to main
... You approved 7+ meaningful* PRs of others
... You did not miss peer reviews for more than one assignment.
Ability to explain lecture contents and defend all aspects of the team project in the oral examination. The inability to explain or justify the project in the oral examination can be repaired in a second oral examination.
Availability of team results for grading (repositories, report, artifacts)

(*) A "meaningful" contribution adds source code, configuration options, or report content. We do not count comments or whitespace.

The full specification of the knock-out criteria will be posted on Brightspace.

--- Summative Assessment ---

You can only get a grade for the course, if you pass the following pass/fail elements:

You have passed all knock-out criteria.
You passed the Presentation & Oral Examination

Your grade then consists of the following partial grades. ALL partial grades are rounded to .1 and must at least be "almost passed" (≥ 5.0). Their weighted average will be rounded to .5 to calculate the final team grade.

Versioning & Releases (10%)
Containers & Orchestration (15%)
Kubernetes & Monitoring (15%)
Provisioning (15%)
Istio Service Mesh (15%)
ML Configuration Management (15%)
ML Testing (15%)

In case of large performance differences in a group, we may choose to assess the team members individually, based on the individual contributions to the deliverables and the interaction traces that we find in the repository.

--- Please Note ---

All assignment deadlines throughout the course are fixed, we do not accept late submissions. Late submissions in systems that do not enforce due dates will be treated as not submitted (e.g., Brightspace, Buddycheck).
Failing to submit a deliverable cannot be repaired.
Insufficient deliverables (≥ 4.0 and < 6.0) can be repaired, but the grade of the deliverable is then capped at 6.0.
There is NO resit opportunity for this course.
Partial grades are not carried over to the next academic year.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5)

CS4570	Machine Learning for Software Engineering	5
Responsible Instructor	M. Izadi	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: - Experience with programming is required (preferably Python) - Experience with machine learning and specifically deep learning techniques is a must-have - Familiarity with transformer-based language models and modern NLP is needed Recommended: - Experience with research methods is nice to have - Taking the NLP course (CS4360) before this class is recommended	
Course Contents	Software repositories archive valuable software engineering data, such as source code, execution traces, historical code changes, mailing lists, and bug reports. This data contains a wealth of information about a projects status and history. Advances in Machine Learning and AI technologies, as demonstrated by the successful application of deep neural networks in various domains did not go unnoticed in the field of Software Engineering; researchers have applied machine learning to tackle different software engineering tasks. This course aims to give students an understanding of a hands-on approach to how deep neural networks and NLP techniques are used to represent knowledge and solve existing SE problems in novel ways.	
Study Goals	At the end of this course, the student is able to: 1. Understand current literature in the area of software analytics and machine learning for software engineering. 2. Apply machine learning / deep learning algorithms to solve software engineering tasks. 3. Apply software analytics techniques to extract actionable software engineering insights. 4. Evaluate model performance on a software engineering task.	
Education Method	Lectures, Seminars and Self-study	
Literature and Study Materials	Research papers	
Assessment	The final grade of the course consists of the following components: - Group Assignments: Project LLM4code report, including code. Propose an idea for improvement of an existing language model for code and implement and assess the same model (weighting 70%) - Group Presentation 1: Project LLM4code presentation: Report and discuss results of the project (weighting 20%) - Individual Presentation: Individual questions and answers about the project LLM4code (weighting 10%) - Group Presentation 2: Present a state-of-the-art approach in the literature (pass/fail) - Group Presentation 3: Present project progress in mid-term (pass/fail) Final grade calculation = $0.7 * \text{Group Assignments} + 0.2 * \text{Group Presentation 1} + 0.1 * \text{Individual Presentation} + \text{Pass/Fail} * \text{Group Presentation 2}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Assignments: Repair opportunity - Group Presentation 1: Resit opportunity - Individual Presentation: Resit opportunity - Group Presentation 2: Resit opportunity - Group Presentation 3: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4575	Sustainable Software Engineering	5
Responsible Instructor	L. Miranda da Cruz	
Instructor	Dr. C.E. Brandt	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Sustainable Software Engineering is an overarching discipline that addresses the long-term consequences of designing, building, and releasing a software project. By definition, sustainability covers five main perspectives: environmental, social, individual, economic, technical. This course mainly focuses on the first, also known as Green Software Engineering. On top of that, we also cover fundamental aspects of social and individual sustainability of software projects. Software Engineering (SE) has long addressed sustainability by narrowing it down to economic and technical sustainability. However, our society is facing major sustainability challenges that can no longer be overlooked by software engineers and computer scientists. It is estimated that, by 2040, the ICT sector will contribute to 14% of the global carbon footprint. Hence, the environmental, social, and individual perspectives ought to be part of the equation when it comes to design, build, and release software systems. The problem is far from simple and we need expert computer scientists to bring sustainability into the core values of the next generation of tech-leading organisations.	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental sustainability principles and their relevance to software engineering. 2. Assess the energy consumption of software systems using measurement and testing techniques. 3. Create actionable strategies to address sustainability issues within software organisations. 4. Evaluate and compare different solutions for sustainable software systems based on their effectiveness in achieving sustainability goals and their potential impact on various stakeholders.	
Education Method	Lectures and Labs	
Literature and Study Materials	Lecture slides and research papers	
Assessment	The final grade of the course consists of the following components: - Group Assignment 1: write a scientific report about AI model energy testing and verification (weighting 40%) - Group Assignment 2: project code and report (weighting 40%) - Group Presentation: project presentation (weighting 10%) - Individual Presentation: individual Q&A after group presentation (weighting 10%) Final grade calculation = $0.4 * \text{Group Assignment 1} + 0.4 * \text{Group Assignment 2} + 0.1 * \text{Group Presentation} + 0.1 * \text{Individual Presentation}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Presentation: Resit opportunity - Individual Presentation: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Visual Computing 2024	
Introduction 1	Visual computing is concerned with theoretical models, algorithms and computational schemes for the acquisition, processing, analysis and synthesis of images and 3D-digital shapes. The recent boom in automatic visual understanding represents an inflection point in human productivity, permeating wide aspects of the economy and society. Examples of visual tasks which are repetitive or require expert knowledge include medical diagnosis, autonomous vehicles/robots, industrial inspection, etc. When machines can meaningfully assist or even completely take-over such visual tasks, it will change the world as we know it. Topics include image formation, multi-scale analysis, style transfer, deepFakes, image/video classification/detection/segmentation, point -clouds, meshes, radiance fields, etc. The theme consists of three courses: Course 1 (Applied Image Processing): Image formation and automatically processing digital images (Image in -> image out). Course 2 (Computer Vision): Extends course 1 to automatic analysis of images and video (images in -> analysis out). Course 3 (3D Visual Computing): Deepens courses 1 and 2 by formation, processing, and analysis of visual 3D data. All courses rely on deep learning. Deep learning replaces feature engineering methods and can successfully learn visual features from huge, annotated datasets. Thus, each course has students implement visual computing algorithms in Python/Pytorch. Every course also has students apply visual computing techniques on a (simple) real-world problem.
Introduction 2	Courses: Q2: DSAIT4120 Applied Image Processing Q3: DSAIT4125 Computer Vision Q4: DSAIT4130 3D Visual Computing

DSAIT4120	Applied Image Processing	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	Dr. M. Weinmann	
Instructor	Dr. P. Kellnhofer	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course provides an overview of image processing with a deeper focus on applications for visual data. This will include: image representations, color models, linear filters, image sampling, digital photography (including HDR and tone mapping), image retargeting, image abstraction and non-photorealism, multiscale image processing (pyramids, Fourier transform), geometrical image transformations (warping, morphing), and neural image processing (concept, texture synthesis, style transfer, super-resolution, denoising, editing).	
Study Goals	At the end of this course, the student is able to: 1. Describe image processing techniques and concepts such as image formation, multi-scale analysis, denoising, super-resolution, style-transfer, texture synthesis, image synthesis, etc. 2. Implement efficient algorithms for processing digital images in C++ or Pytorch. 3. Apply image processing algorithms to a (simple) real-world problem which may need to run, evaluate, or extend existing code. 4. Analyse algorithms for image processing based on their computational costs, qualitative performance, and application areas. 5. Reflect on an image processing/editing/manipulation assignment in the context of responsible engineering and ethics.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of image processing, common problems and their solutions. Various algorithms and their properties will be discussed. During the lab sessions (2 hours each), you will be introduced to guidelines for efficient implementation of selected algorithms, and you will receive formative feedback on your approach to practical assignments. The final project will be solved in small groups, and it will focus on survey and assessment of state-of-the-art approaches to a specified image processing problem. It will be closed by a presentation and a report.	
Literature and Study Materials	Required literature: - Lecture slides with additional references. Recommended literature: - Szeliski, Richard. Computer vision: algorithms and applications. Springer Science & Business Media, 2010. - Chapters 3, 4 and 10. - Burger, Wilhelm, and Mark J. Burge. Digital image processing: an algorithmic introduction using Java. Springer, 2016. - Chapters 22-26. - Sundararajan, Duraisamy. Digital image processing: a signal processing and algorithmic approach. Springer, 2017. - Chapters 2-7. - Goodfellow, I., Bengio, Y. and Courville, A.. Deep Learning. MIT Press, 2016. - Chapters 6-9 and 14.	
Assessment	The final grade of the course consists of the following components: - Individual Assignments (weighting 55%) - Group Assignment (weighting 45%) Final grade calculation = 0.55 * Individual Assignments + 0.45 Group Assignment A passing final grade for the course can only be earned if both components are at least a 5.0 and the weighted final grade (rounded to 0.5) is at least a 6. The group assignment can lead to individual grades and not necessarily a single grade for all group members. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Repairs are individual and the group assignment can only be repaired, if the individual assignments (without repair) was superior or equal to a 6.0. - Individual Assignments: Repair opportunity via an oral exam with potential preparation time. - Group Assignment: Repair opportunity in form of an individual project. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4125	Computer Vision	5
Responsible Instructor	Dr. J.C. van Gemert	
Responsible Instructor	N. Tömen	
Instructor	Dr. X. Zhang	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course is on automatically understanding visual content such as images and videos by deep learning. The range of topics (although, not limited to): fundamentals in vision, visual representations, object detection, per-pixel labelings, video recognition, image similarity learning, efficiency, self-supervision, etc.	
Study Goals	At the end of this course, the student is able to: 1. Describe computer vision techniques for images and video such as visual representations, classification, detection, segmentation, optical flow, action recognition, retrieval, etc.. 2. Implement computer vision techniques in Python/Pytorch. 3. Identify the core contributions of a computer vision research paper. 4. Debate related literature for a given computer vision research paper. 5. Apply appropriate computer vision techniques to a (simple) real-world problem which may need to re-implement, run, evaluate, investigate, extend existing research or code. 6. Write clearly and concisely about your motivation, code, method, results, and analysis.	
Education Method	The course will have lectures, a seminar and a lab practical: - The lectures will be on established topics based on the current literature. - The seminar will have students read, critique, present, and debate relevant computer vision research papers. You will have to read 2 papers per week, for 7 weeks (14 papers). - The lab will have students apply and design their own (small) computer vision project.	
Literature and Study Materials	S. Prince; Understanding deep learning, https://udlbook.github.io/udlbook/	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 50%) - Group Report: blog post about the group project (weighting 40%) - Group Presentation: presentation about a research paper (weighting 10%) Final grade calculation = $0.5 * \text{Written Exam} + 0.4 * \text{Group Report} + 0.1 * \text{Group Presentation}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Report: Repair opportunity - Group Presentation: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4130	3D Visual Computing	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	Dr. P. Kellnhofer	
Instructor	Dr. K.A. Hildebrandt	
Instructor	Dr. M. Weinmann	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course is concerned with 3D visual computing techniques and concepts such as point clouds, meshes, 3D capturing, structure from motion, shape-from-X, multi-view reconstruction, shape analysis, geometric deep learning, radiance fields, image-based rendering etc.	
Study Goals	At the end of this course, the student is able to: 1. Describe 3D visual computing techniques and concepts such as point clouds, meshes, 3D capturing, structure from motion, shape-from-X, multi-view reconstruction, shape analysis, geometric deep learning, radiance fields, image-based rendering etc. 2. Implement 3D processing techniques and concepts in Python/Pytorch. 3. Apply appropriate 3D visual computing techniques to a (simple) real-world problem which may need to run, evaluate, or extend existing code. 4. Debate related literature for a given 3D Visual Computing technique.	
Education Method	The course consists of two types of weekly contact activities. In the lectures, you will be presented with the theory of 3D Visual Computing, common problems and their solutions. Various algorithms and their properties will be discussed. During the guided lab sessions, you will have a chance to work on your programming assignments.	
Literature and Study Materials	Course slides and research papers	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 50%) - Individual Assignments: individual programming assignments (weighting 50%) Final grade calculation = 0.5 * Digital Exam + 0.5 * Individual Assignments A passing final grade for the course can only be earned if all components have at least a grade of 5.0 and the weighted average rounded on 0.5 is at least a 6. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Individual Assignments: A (potentially group) replacement project Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	2024/2025
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Data Science and Artificial Intelligence Technology

Research courses 2024	
Introduction 1	In the fourth quarter, you are prepared to conduct research in DSAIT. You learn state-of-the-art research practices to examine DSAIT knowledge critically. For this purpose, you choose one of the research courses from the list below. The choice of research course allows you to differentiate between learning general research skills for data science and AI technology or learning these skills for a specific field. During the course, you will learn in-depth research practices and work with integrity and responsibility in a concrete DSAIT setting. You will build on the knowledge gained in the DSAIT themes.
Introduction 2	Courses: DSAIT4200 Empirical Research of Computational Solutions DSAIT4205 Fundamental Research in Machine and Deep Learning DSAIT4210 Research in Web Data and Information Management DSAIT4215 Research in Visual Computing and Interactive Visual Analysis DSAIT4220 Research in Intelligent Decision Making DSAIT4225 Research in Bioinformatics DSAIT4230 Research in Social Signal Processing and Affective Computing DSAIT4235 Research Seminar on Multimedia Computing and Systems

CESE4065	Advanced Practical I.o.T. and Seminar	5
Responsible Instructor	Dr. R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	2/0/0/0 and 0/0/0/2	
Education Period	1 4	
Start Education	1 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.	
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break	
Summary	his course is an involved hands-on course for self-motivated students. Students usually work in a group of two (max 3). Students are expected to have sufficient programming and hardware development skills. The course is project-oriented thus students are expected to deliver a working model with demonstrations. ----- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. ----- This course is done in a group of two (max 3). The instructor/TA will help connect a potential groupmate in the first class using Brightspace/Feedback Fruits. ----- Activities: How to in Groups? ----- Part 1: Seminar We guide the groups in selecting a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar, all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID-19-related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to the 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/Raspberry Pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki-based networks or Open testbed projects. In these cases, online group activities are possible. NOTE: In some extreme cases, the Instructor/TA may allow one person to project after assessing the situation and after discussing it with the student. ----- Supply of Hardware: ----- 1. Instructor/TA is able to provide take-home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components. Course Contents Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers. Study Goals To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature. Education Method Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s). Literature and Study Materials This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.	

Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc. Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project. Part 1: 30% of the whole marks; Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment. Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference. Report will not have individual component, however, Q/A may have individual component. Please NOTE: There would be no resit since this is a hands-on course being executed in a team. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
Tags	Circuits Group work Programming Project Software
maximum aantal deelnemers	Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

DSAIT4200	Empirical Research of Computational Solutions	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Responsible Instructor	J. Urbano Merino	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Recommended: Basic programming skills (e.g., CSE1100), Probability Theory and statistics (e.g., CSE1210)	
Course Contents	In today's world, where an increasing number of computational solutions like recommender systems, social robots, and health monitoring and wearable devices are being introduced, empirical research plays a crucial role in driving innovation, uncovering insights, and making informed decisions. This course equips students with fundamental skills in conducting empirical research to evaluate these computational solutions and their underlying theories. The course begins by providing students with a solid understanding of empirical research methods. They learn to set up a research study, collect and analyse data, and draw scientifically valid conclusions. Emphasis is placed on reproducible research practices, including techniques like pre-registration and creating a data plan. To make sense of data samples, students study how to make statistical inferences about the population. They explore frequentist and Bayesian data analysis approaches, gaining the ability to make meaningful statistical inferences based on collected data. Throughout the course, students work with computational tools that aid in conducting statistical analyses of real-world data. Using these tools, students gain practical experience analysing and interpreting data to draw meaningful conclusions. By the end of this course, students will have a strong foundation in empirical research methods and the ability to evaluate computational solutions. They will be equipped with practical skills in data analysis, statistical inference, and working with real-world datasets. The course aims to empower students to become proficient researchers who can contribute to advancements in computational solutions and their underlying theories. The course's main topics: - Conceptualizing research questions and experimental design, and data planning - Frequentist and Bayesian data analysis - Generalized linear models for statistical analysis - Multilevel modelling for hierarchical and longitudinal data analysis - Measuring and sampling, validity and reliability - Principles of statistical testing	
Study Goals	At the end of this course, the student is able to: 1. Compose a scientifically sound research plan for an empirical study of a computational solution 2. Make statistical inferences with computational tools based on the Frequentist approach to data analysis 3. Make statistical inferences with computational tools based on the Bayesian approach to data analysis 4. Formulate conclusions that can be drawn from statistical inferences about computational solutions and their underlying theories 5. Address typical biases and misunderstandings in empirical research	
Education Method	In the lectures, theories, principles and methods are presented, discussed and illustrated with examples. Outside the lectures, students work in small groups on their assignments using R.	
Literature and Study Materials	- McElreath, R. (2020). Statistical rethinking: A Bayesian course with examples in R and Stan. CRC press. - Robson, C., McCartan K. (2016) Real World Research. John Wiley & Sons - Field, Z., Miles, J., & Field, A. (2012). Discovering statistics using R. Discovering Statistics Using R, 1-992.	
Assessment	The final grade of the course consists of the following components: - Group Presentation: presentation of pre-registration of empirical study (including a data plan) and the results of data analysis assignment based on the Bayesian approach (weighting 35%) - Individual Presentation: individual Q&A session about the pre-registration of empirical study (including a data plan) and the results of data analysis assignment based on the Bayesian approach (weighting 15%) - Group Report: report on the results of data analysis assignment based on a frequentist approach (weighting 50%) Final grade calculation = $0.35 * \text{Group Presentation} + 0.15 * \text{Individual Presentation} + 0.5 * \text{Group Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Presentation: Resit opportunity - Individual Presentation: Resit opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 9, sub 5).	

DSAIT4205	Fundamental Research in Machine and Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Responsible Instructor	M.A. Migut	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is about doing fundamental research in machine/deep learning. It is about doing research to better understand ML/DL models, what assumptions they make, when they fail, how to pose research questions, answer empirical hypotheses experimentally and analyse results. Machine/deep learning research goes hand-in-hand with designing applications, superior performance, more data, improved accuracy, faster inference, bigger models, better algorithms, solving a problem, creating a new artifact/tool/AI, etc. While these concepts are all valuable in their own right, they are NOT the topic of this course. This course is NOT about applying ML/DL to "solve" some problem. Instead, here, we study how to "understand" and do fundamental research in ML/DL itself. This course is about repeatedly asking "Why?"; it's about better understanding, reflecting, questioning, reproducing, and analysing ML/DL research. Students will read, present and debate scientific papers on machine learning scholarship. Construct a logical research "storyline", create synthetic toy problems, reproduce existing research, while honing a critical attitude and being able to communicate concisely and clearly.	
Study Goals	At the end of this course, the student is able to: 1. Debate related scientific papers on research in ML/DL. 2. Construct a concise, complete, logical, and self-contained motivation (storyline) for a research project including why the topic is relevant, how its done now, problem statement/research question, empirical/experimental questions, etc. 3. Construct a synthetic, fully controlled experimental setup that tests a single hypothesis of a ML/DL paper. 4. Reproduce (part of) an existing ML/DL paper which may require a re-implementation, evaluation, investigation, SWOT analysis, etc. 5. Critically assess the scientific value of a ML/DL paper, including: confusing explanation with speculation, failure to identify the sources of empirical gains, identifying failure cases, etc. 6. Write clearly and concisely about your storyline, motivation, code, approach, results, analysis and reflection.	
Education Method	In the seminars, students present and discuss research papers. In the lectures, we plenary discuss, reflect, answer Q&As In the labs, students get supervision on construct the storyline, toy problems, and the reproduction.	
Literature and Study Materials	Parts of the research guidelines, available via: https://jvgemert.github.io/links.html	
Assessment	The final grade of the course consists of the following components: - Group Presentation: presentation about an assigned paper on research methodology (weighting 10%) - Individual Report 1: storyline (weighting 15%) - Individual Report 2: toy problem (weighting 15%) - Group Report: reproduction blog post about reproducing (a part) of a ML/DL paper, each student has an individual part in the reproduction (weighting 60%) Final grade calculation = $0.1 * \text{Group Presentation} + 0.15 * \text{Individual Report 1} + 0.15 * \text{Individual Report 2} + 0.6 * \text{Group Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Presentation: Resit opportunity - Individual Report 1: Repair opportunity - Individual Report 2: Repair opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4210	Research in Web Data and Information Management	5
Responsible Instructor	Prof.dr.ir. G.J.P.M. Houben	
Responsible Instructor	Dr. M.S. Pera	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: BSc Computer Science (or equivalent)	
Course Contents	In this course we discuss recent developments in the area of web data and information management. We select topics to discuss from the areas of: - web technology (e.g., web engineering, hypertext, adaptive web), - web data management (e.g., web data interoperability, system and data integration), - web data and semantics (e.g., ontologies, semantic web, metadata), - web data analytics (e.g., user modeling, web personalization, web information filtering and retrieval), - social web (e.g., social web data analytics, social networking, human computing), - web science (e.g., crowdsourcing, trust, data science). We discuss this content while learning about the role of scientific communication and about the scientific methodologies and approaches for conducting research in the area. In this research course the students will have to prepare and give scientific presentations on the basis of research papers about selected topics - the topics are selected in the first session together with the students. The students will also have to attend the presentations and participate in discussions on the presented papers. In addition, students will have to write a short survey about a topic in the area of web data and information management of their choice.	
Study Goals	At the end of this course, the student is able to: 1. Explain the current developments in research on web information systems, the methodologies and approaches to conduct research in the area; 2. Analyse literature relevant to a given problem 3. Critically assess scientific literature, including the importance of the academic journals and conferences in the area and their review process. 4. Create a research plan 5. Write the draft of a research paper guided by the research plan	
Education Method	Seminar, with discussions, presentations, feedback sessions, and 1on1 Mentoring	
Literature and Study Materials	Is provided in the research course depending on the chosen subjects.	
Assessment	The final grade of the course consists of the following components: - Individual Presentation: presentation discussion state of the art literature (weighting 15%) - Individual Report: report outlining a research plan and research proposal (weighting 75%) - Group Presentation: group discussion in class (weighting 10%) Final grade calculation = $0.15 * \text{Individual Presentation} + 0.75 * \text{Individual Report} + 0.1 * \text{Group Presentation}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Presentation: Repair opportunity - Individual Report: Repair opportunity - Group Presentation: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4215	Research in Visual Computing and Interactive Visual Analysis	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	Dr. P. Kellnhofer	
Instructor	Dr.ir. A.R. Bidarra	
Instructor	Dr. K.A. Hildebrandt	
Instructor	T. Höllt	
Instructor	Dr. R. Marroquim	
Instructor	Dr. M. Skrodzki	
Instructor	Dr. M. Weinmann	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Recommended: DSAIT Theme Visual Computing or Interactive Visual Analysis	
Course Contents	In this research course you conduct deep investigation of recent techniques solving attractive problems in Visual Computing and Interactive Visual Analysis including Visualization, Interaction, Image Processing, Inverse rendering and 3D data processing. You will survey research outputs in the area of your interest, present a selected paper and develop an implementation supporting the education on the topic. The goal is to strengthen you analytical and presentation skills and prepare you for research in your MSc project.	
Study Goals	At the end of this course, the student is able to: 1. Explain a representative method covering selected topics from the field of Visual Computing and Interactive Visual Analysis in form of a public lecture. 2. Formulate insightful questions during presentations of others. 3. Implement a tool that helps to further explains the discussed technique. 4. Critically assess results of the implementation in form of a short report and a presentation.	
Education Method	This course has the format of a student seminar. You will prepare a lecture on a selected topic from Visual Computing and Interactive Visual Analysis. Typically, you will present a detailed look at one representative method based on a scientific paper. The presentation goes in-depth and covers the strengths and weaknesses of the described technique. Depending on number of participants, we may organize the paper presentations individually or in groups. After each presentation in the plenary colloquium session, a research discussion takes place. All students participate in a scientific discussion and ask relevant critical questions. Finally, you will implement a tool that further explores the presented method. Depending on the number of participants and scheduling constraints, you will then be asked to present your work either in a form of a short public demonstration or in a pre-recorded video presentation. Additionally, you will prepare a short report summarizing your solution.	
Literature and Study Materials	Scientific journals and conference proceedings from conferences including SIGGRAPH, Eurographics, EuroVis, CVPR, ICCV, ECCV and similar.	
Assessment	The final grade of the course consists of the following components: - Presentation of a recent scientific paper (50%) - Implementation component reproducing an aspect of the paper (40%, including short report and presentation) - Preparation of questions for the presented research papers (10%) Final grade calculation: (0.5 * Presentation + 0.4 * Implementation + 0.1 * Questions) A passing final grade for the course can only be earned if all components have at least a grade of 5.0 and the weighted average rounded on 0.5 is at least a 6. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Presentation of a recent scientific paper can be repaired by an individual video presentation of a state-of-the-art report + an oral examination. The content of each presented paper must be understood by the student. - Implementation component can be repaired by a new implementation following a new approved plan. - Preparation of each question for the assigned paper can be repaired by a short video presentation of the same paper.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4220	Research in Intelligent Decision Making	5
Responsible Instructor	S. Dumani	
Responsible Instructor	Dr. J.W. Böhrmer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	ONLY OPEN FOR STUDENTS FROM M-CS AND M-DSAIT THAT FOLLOW THEMES: - M-CS Theme Algorithmics - M-DSAIT Theme Optimisation & Reasoning - M-DSAIT Theme Probabilistic Decision Making	
Course Contents	Intelligent decision making is a key skill of computational agents. Research on this topic focuses on building models and algorithms that enable AI systems to take appropriate decisions. Building upon theoretical knowledge gained in the Probabilistic Decision Making or Reasoning and Search themes, students collaborate in small groups on a distinct research project per group, for instance on decision-making problems in transport, logistics or smart energy grids. Purely algorithmic challenges will also be provided.	
Study Goals	The research projects provide a good opportunity to learn about topics suitable for Masters projects. At the end of this course, the student is able to: 1. Apply algorithms for decision making to problem domain for comparison and evaluation. 2. Create an extension of a decision-making algorithm. 3. Critically assess relevant topics in the research field of algorithms for intelligent decision making. 4. Apply the appropriate research methodology. 5. Communicate their findings effectively.	
Education Method	A research project in a small group	
Literature and Study Materials	Research papers	
Assessment	The final grade of the course consists of the following components: - Group Assignment: quality of work of the research project (weighting 40%) - Group Report: a scientific report of the research project (weighting 20%) - Individual Assignment: performance during the project (weighting 30%) - Group Presentation: presentation of the research project (weighting 10%) Final grade calculation = $0.4 * \text{Group Assignment} + 0.2 * \text{Group Report} + 0.3 * \text{Individual Assignment} + 0.1 * \text{Group Presentation}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Assignment: Repair opportunity - Group Report: Repair opportunity - Individual Assignment: No - Group Presentation: Resit opportunity	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). Prof.dr. M.T.J. Spaan	

DSAIT4225	Research in Bioinformatics	5
Responsible Instructor	Prof.dr.ir. M.J.T. Reinders	
Responsible Instructor	Dr. T.E.P.M.F. Abeel	
Instructor	Dr. J.S. de Pinho Gonçalves	
Instructor	Dr. J.A. Baaijens	
Instructor	J.M. Weber	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	ONLY OPEN FOR M-DSAIT STUDENTS FOLLOWING THE BIOINFORMATICS THEME	
Course Contents	Bioinformatics is at the heart of many modern systems biology analyses, and encompasses the application of statistics and computer science to (large-scale) biomolecular datasets. In essence, bioinformatics is about smart ways of extracting knowledge from the enormous amounts of data that can be generated using modern measurement techniques. For instance, it plays an important role in finding the genetic origins of various diseases, such as cancer, diabetes or Alzheimer. In this course we study some key examples of relevant data-driven bioinformatics methods by reading and discussing a set of selected papers that present interesting DSAIT solutions with significant biological conclusions.	
Study Goals	At the end of the course, the student is able to: 1. Explain common high-throughput bioinformatics data types, such as DNA/RNA sequencing, along with experiments for their acquisition. Discuss the benefits and limitations of these methods. 2. Discuss data-driven bioinformatics problem formulations in the literature, and their relationship with real-world challenges, assessing the alignment between them and identifying assumptions inherent to the problem definition. 3. Explain and execute solutions for data-driven bioinformatics problems (methods, algorithms, and models) in own words to both experts and non-experts based on descriptions provided in the literature. 4. Reason about and argue on the merits and limitations of solutions proposed for data-driven bioinformatics problems concerning both general DSAIT theory and specific alignment with the problem formulation. 5. Assess and debate the evaluation strategy accompanying solutions for data-driven bioinformatics problems proposed in the literature. Identify merits and potential caveats of such evaluation strategy. 6. Communicate effectively on the understanding and critical reflection of data-driven bioinformatics literature.	
Education Method	The course uses a flipped class-room set-up, where students take turns presenting on the selected papers. In each lecture (2h), one of the selected papers will be discussed in detail. Lecturers moderate the discussion and fill in any gaps in understanding. All students are required to read, and prepare the course material before every lecture to effectively participate. Each student should be prepared to raise at least two critical remarks or questions during the discussion.	
Literature and Study Materials	Papers selected from relevant Bioinformatics-related journals.	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 60%) - Individual Presentation: presentation explaining, analysing, and discussing a particular scientific paper (weighting 20%) - Individual Assignment: raise and discuss at least two critical remarks or questions (weighting 20%) Final grade calculation = $0.6 * \text{Digital Exam} + 0.2 * \text{Individual Presentation} + 0.2 * \text{Individual Assignment}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Individual Presentation: Repair opportunity - Individual Assignment: No Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4230	Research in Social Signal Processing and Affective Computing	5
Responsible Instructor	Dr. H.S. Hung	
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Instructor	Dr. B.J.W. Dudzik	
Instructor	Dr. C.A. Raman	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The core of social and emotional intelligence is our ability to understand and interpret social and affective signals of a person we are communicating with is. Social-emotional intelligence is a facet of human intelligence that has been argued to be indispensable and perhaps the most important for success in life. Social Signal Processing and Affective Computing (SSPAC), aims to develop automated approaches that can interpret human social and/or affective behaviour through machine analysis and production of nonverbal and verbal behavior. In this course you will learn how next-generation computing can make use of such social and affective signals by giving it the ability to recognize and produce human social signals and social behaviors. Think about turn taking, politeness, disagreement, emotions, rapport. You will learn about relevant findings in social psychology, and you will learn computational techniques that allow systems to make use of social and affective signals to become more effective and more efficient by being able to detect but also simulate (e.g. in virtual agents) blinks, smiles, crossed arms, laughter. Socially aware computing. These techniques can be used in robots, virtual agents, smart homes, crowd monitoring, etc.	
Study Goals	At the end of this course, the student is able to: 1. Describe what social and affective signals are. 2. Apply computational methods to detect and simulate social and/or affective signals. 3. Position the field of social signal processing and affective computing in computer science and psychology, and identify its major goals and angles of study. 4. Critique, major psychological theories of social interaction. 5. Explain and critique major social and affective signal recognition, simulation and expression techniques in computational systems. 6. Develop and motivate a research project that uses social and/or affective signals in a non-trivial manner with hypotheses, research questions, and a supporting literature survey. 7. Implement and evaluate (in groups) a research project that uses social and/or affective signals in a non-trivial manner with hypotheses, research questions, and a supporting literature survey.	
Education Method	Seminar: - Crash Course in Social Signal Processing and Affective Computing - Self-study of papers and book. - The papers will be made available at the start of each lecture. Project: Perform a piece of research (survey, research question, programming, testing) and write a paper about it. Students will work in teams of about 3-4 people	
Assessment	The final grade of the course consists of the following components: - Group Presentation: midterm presentation (weighting 10%) - Group Report 1: proposal project (weighting 30%) - Group Report 2: final project report (weighting 40%) - Individual Presentation: presentation final project (weighting 20%) Final grade calculation = $0.1 * \text{Group Presentation} + 0.3 \text{ Group Report 1} + 0.4 * \text{Group Report 2} + 0.2 * \text{Individual Presentation}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Presentation: Resit opportunity - Group Report 1: Repair opportunity - Group Report 2: Repair opportunity - Individual Presentation: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4235	Research Seminar on Multimedia Computing and Systems	5
Responsible Instructor	P.S. Cesar Garcia	
Responsible Instructor	M. Mansoury	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Through all the exciting recent advances in digital media technology and the rapid growth of social media platforms, multimedia content is increasingly embedded in our daily lives, gaining enormous potential in improving the traditional educational, professional, business, communication and entertainment processes. To be able to use this potential for transferring these processes into user-centric interactive multimedia applications, technology is required that can help us access, deliver, enrich and share rich-media content. This course provides insight into the state-of-the-art cross-disciplinary research efforts related to the development of such technology. The topics covered by the course include, but are not limited to, multimedia engagement (emotional and social signals and recommender systems), multimedia technology (language and speech technology, telepresence and VR, mobile), and multimedia experiences (Quality of Experience, collaboration), and Multimedia in the Generative AI Era.	
Study Goals	At the end of the course, the student is able to: 1. Understand the state-of-the-art research and development activities in the field of Multimedia Computing and systems 2. identify the "knowledge gap" (i.e., the place in which more research is needed in order to advance the state of the art). 3. write a survey paper on one specific topic 4. Present this survey paper to an audience	
Education Method	Readings, seminar discussions, presentations, survey paper	
Literature and Study Materials	Papers and articles	
Assessment	The final grade of the course consists of the following components: - Individual Report: write own paper/article (weighting 65%) - Individual Presentation 1: presentation pre-existing paper (weighting 10%) - Individual Presentation 2: present own survey (weighting 25%) Final grade calculation = $0.65 * \text{Individual Report} + 0.1 * \text{Individual Presentation 1} + 0.25 * \text{Individual Presentation 2}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Report: Repair opportunity - Individual Presentation 1: Resit opportunity - Individual Presentation 2: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29 sub 5).	

Year	2024/2025
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Data Science and Artificial Intelligence Technology

Deepening/broadening Electives 2024

Year	2024/2025
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Data Science and Artificial Intelligence Technology

Interdisciplinary projects

IFEEMCS520201	Advanced Interdisciplinary Artificial Intelligence Project	15
Responsible Instructor	J. Zatarain Salazar	
Responsible Instructor	Dr. P.K. Murukannaiah	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	FROM SEPTEMBER 2025, THIS COURSE WILL BE PART OF IFM4040 JOINT INTERDISCIPLINARY PROJECT (JIP) AND WILL NO LONGER BE OFFERED UNDER THIS COURSE CODE. ENROLLMENT FOR THE COURSE STARTS MARCH 17TH 2025 VIA BRIGHTSPACE (IFM4040) This course is aimed at MSc students from all TU Delft faculties who have already obtained AI-related knowledge and skills in previous courses, and who want to apply their skills to a real-world AI challenge. Prospective students, for example, will have completed an AI minor or have a good amount of AI knowledge from their bachelor (e.g. Mechanical Engineering), on topics such as data handling and machine learning OR possess demonstrated expertise in the domain relevant to a specific AI project on offer. This is an indicative guideline and we ask students that want to enroll for this course, to send a motivation letter and give an indication of their experience and knowledge on AI.	
Course Contents	FROM SEPTEMBER 2025, THIS COURSE WILL BE PART OF IFM4040 JOINT INTERDISCIPLINARY PROJECT (JIP) AND WILL NO LONGER BE OFFERED UNDER THIS COURSE CODE. ENROLLMENT FOR THE COURSE STARTS MARCH 17TH 2025 VIA BRIGHTSPACE (IFM4040) Artificial Intelligence (AI) is an interdisciplinary challenge. This Advanced Interdisciplinary Artificial Intelligence Project (AI2P) (IFEEMCS520201) is a 15EC MSc elective course aims to provide students with a unique experience to learn, develop and apply AI to a real-world challenge, working in an interdisciplinary project team. In this course you work with a group (4-6 students) on a large interdisciplinary project that is run by TU Delft AI Labs. [link: https://www.tudelft.nl/ai/tu-delft-ai-labs] Further, some of the projects may offer business cases from the industry. You and your fellow students will be offered to join workshops, tutorials and seminars organized by cutting edge researchers from TU Delft. You will also share kick-offs, reviews, and events with students from JIP (Joint Interdisciplinary Project). You find the projects for next academic here on Project Forum [link: https://projectforum.tudelft.nl/course_editions/109]	
Study Goals	Upon finishing this course, the student should be able to: 1. position an AI challenge in the broader literature, e.g., by performing a literature review on the AI challenge in a domain; 2. set-up a research strategy for an interdisciplinary AI project; 3. develop the AI solution according to the research strategy; 4. effectively collaborate with the team and supervisor; 5. evaluate and reflect on the research / developed AI solution; 6. assess the impact of deploying AI-based solutions and interventions on individuals, organizations, and society; and 7. report and present their research on a scientific level.	
Education Method	The students may work and discuss in groups. Each student is responsible for gathering, sharing and implementing information within the overall project. This allows for collaboration and communication as well as self-directed learning. At least once a week, the students meet with their supervisor. Peer feedback as well as feedback from the supervisor is organized on essential steps in the research project, which include: 1. Project proposal 2. Mid-term review, and 3. A scientific report, describing the problem, methods, results, and conclusion 4. A scientific presentation	
Assessment	The scientific report and the project presentation are assessed by the project supervisor and an external examiner. The assessment based on the clarity of presentation, the soundness of the methods used, and the conclusions drawn. Resit next quarter: A repair option is allowed only for students that received a fail mark (<5.8). Overall, the resit mark will be capped to 5.8.	

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should - be a 2nd-year master student; - have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. - be full time available (40 hours per week) in Q5; - meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman, VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (ILO: K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

The minimum grade for the summative assessment is ≥ 5.8 .

Process grade is 20%

The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

TPM423A	Technology Venture Development	15
Module Manager	Dr. A. Giga	
Instructor	Dr. A. Giga	
Instructor	J. Schmutzler	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	Join us in an interdisciplinary exploration of technology startups business model and strategy. This pioneering course offers you an opportunity to work with a technology-based startup from the local ecosystem or a corporate venture in an internship-like setting and help them progress. In a group of three to four students from different faculties and study programs, you will work on two projects: 1) Validate and challenge startups business model and strategy 2) Provide consulting on a specific business challenge provided by the startup. In addition, you will be coached and taught solid business theoretical foundation and practical lessons. You will apply your learnings directly, solving pressing business challenges and driving innovation. This hands-on experience allows to you communicate directly with startups key decision-makers and see the effects of your work making a difference. This course is a real-world expedition into business model innovation, designed to inspire, challenge, and prepare you for a future where you drive change. We expect you to spend around three days a week with the startup and about one day a week in the classroom setting. Given its 15 ECTS, we expect your full weekly commitment, as in any joint interdisciplinary project. The applications opens on April 30th and the application deadline will be on June 15th. Here is the link to the application form: https://forms.office.com/e/tX9hXqNM7u	
Course Contents	What to Expect Internship-like Experience: Step into the fast-paced world of startup or corporate venture, through our internship-like experience. Here, you'll gain invaluable hands-on exposure to real-world business practices, navigating challenges, and seizing opportunities alongside industry professionals. Lectures: Immerse yourself in captivating lectures that serve as the cornerstone of your theoretical understanding. Delivered by experts in the field, these sessions will unravel the intricacies of business model innovation, offering profound insights and cutting-edge perspectives. Each week delves into critical topics such as market dynamics, design thinking, team development, financial strategies, risk management, and growth mechanisms, ensuring you're equipped with a comprehensive toolkit for the entrepreneurial world. Sessions illuminate the intersection between technical innovation and business strategy, enriching your learning experience and equipping you with versatile skills for multifaceted business environments. Coaching Sessions: These sessions help you apply your newfound knowledge and skills to your cases and tackle pressing business challenges head-on, devising innovative solutions that drive tangible impact and transformation. What You'll Experience and Learn Interdisciplinary Exploration: Navigate the complex nexus of diverse disciplinary backgrounds in engineering, basic science and architecture and design, entrepreneurship, and management, transcending traditional disciplinary boundaries to craft innovative business models that redefine the status quo. Practical Application: Move beyond theoretical concepts and immerse yourself in practical, hands-on experiences. From market analysis to financial modelling, you'll apply cutting-edge techniques and frameworks to real-world scenarios, honing your problem-solving skills and business acumen. Innovation Mindset: Cultivate an entrepreneurial mindset rooted in creativity, resilience, and adaptability. Through design thinking workshops, risk management exercises, and growth strategy simulations, you'll embrace uncertainty as an opportunity for growth and innovation. Networking Opportunities: Forge meaningful connections with industry professionals, guest lecturers, and fellow classmates, expanding your professional network and gaining valuable insights into the startup ecosystem. Personal and Professional Growth: Reflect on your journey, identify your strengths, and chart a course for future success. Whether it's sharpening your communication skills or refining your leadership abilities, this course offers a transformative experience that extends far beyond the classroom. Why Choose This Elective Cutting-Edge Curriculum: Stay ahead of the curve with a curriculum that reflects the latest trends, technologies, and best practices in business model innovation. Hands-On Experience: Gain practical, real-world experience that sets you apart in today's competitive job market. Our internship-like experience and consulting project offer unparalleled opportunities for skill development and professional growth. Inspiring Faculty: Learn from esteemed faculty members who are experts in their fields and passionate about empowering the next generation of innovators and entrepreneurs. Impactful Learning: Make a meaningful impact with a course that challenges you to think critically, solve complex problems, and drive positive change in the world.	
Study Goals	After completing this course you will have learned to: - Identify and innovatively transform a business idea for a technology venture. - Conduct rigorous market analysis and validation using primary and secondary research methods. - Construct a detailed and financially viable business proposition. - Apply niche strategies to commercialize radical high-tech products. - Come up with innovative, viable, high quality recommendations that will solve real-life challenges for an existing venture, under	

	considerable time constraints. Employ relevant academic theories to support your recommendations; Effectively communicate your product and business proposition to diverse stakeholders. Emulate effective entrepreneurial behaviour in decision-making under uncertainty. Reflect on the evolution of entrepreneurial growth and business model adaptation.
Education Method	This course will entail a mix of lectures, and guest lectures. You will be provided with course materials. You will go through an onboarding process emulating a traineeship experience with the startup or corporate venture. There will be coaching sessions with experienced mentors while will offer guidelines, feedback, and tools to help you develop a solution to the consulting project.
Prerequisites	There are no prerequisites for this course.
Assessment	Your grade is determined as follows: 80% is group work and based on the quality of your group project. Quality is assessed based on your problem statement & presentation, midterm report & presentation and final report and presentation. Startup team will have a strong say on your performance. Adjustment is made through peer evaluation of your individual contribution to the team work. 20% is an individual grade based your personal learning and team reflections during the course.
Enrolment / Application	This course is open to master students across all TUD faculties starting their second year. MAXIMUM CAPACITY is contingent upon the capacity of our partners (startups and corporates) but initially limited to 30 participants. Please apply for the course no later than June 15th. If you would like to apply , fill out the application form: https://forms.office.com/e/tX9hXqNM7u If you are having trouble accessing the form, need help approving this course as your master programs elective, or have any other questions, please write to DCE@tudelft.nl If more than 30 participants apply for the course, we will select based on your application. If you apply after June 15th, you may still join if there is space available.
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.
Elective	Yes
Tags	Business Economics Finance
Targetgroup	MSc students from all faculties can apply.

Year	2024/2025
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Data Science and Artificial Intelligence Technology

Deepening electives

DSAIT4300	Formal Methods for Machine Learning	5
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 This course will cover several topics at the intersection of formal methods and machine learning, for instance, abstraction refinement for analyzing deep neural networks, algorithms for reliable learning-based control, interpretable model design, and formal methods for verification and monitoring of learned systems. The course should be useful for students of both formal methods and machine learning, and lies at the intersection of these areas. Since the area is relatively new, the course material will be primarily based on research papers.	
Study Goals	At the end of this course, the student is able to: 1. modeling complex systems, 2. formally defining critical properties, 3. applying verification tools to learned systems, 4. synthesizing controllers for learned systems, 5. providing interpretability for learned models.	
Education Method	The course will consist of one lecture and one discussion session per week. The students will be assigned projects in groups of two, where they will need to apply the material from the lectures, e.g., train a learned system, define a property and verify it using one of the open-source tools from state-of-the-art research papers. Discussion sessions will be used for questions about the projects.	
Assessment	The final grade of the course consists of the following components: - Participation (weighting 15%) - Presenting Papers (weighting 15%) - Assignment #1 (weighting 15%) - Assignment #2 (weighting 15%) - Final Report (weighting 40%) Final grade calculation = $0.15 * \text{Participation} + 0.15 * \text{Presenting Papers} + 0.15 * \text{Assignment \#1} + 0.15 * \text{Assignment \#2} + 0.4 * \text{Final report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Presenting Papers: resit - Assignment #1: re-submit deliverable - Assignment #2: re-submit deliverable - Final report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4305	Machine Learning for Graph Data	5
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 Graph data are present in a myriad of modern computer sciences systems and applications. Examples include data generated over social, brain, financial, power, water and sensor networks. Because these data have a complicated structure they require different tools conventionally used in machine and deep learning to develop end to end solutions. These solutions falls generally under the umbrella of graph-based machine learning and can be used to perform recommendations, detect anomalies in the brain, predict financial crisis, estimate the state of a power or water network, and coordinate group of autonomous moving sensor, to name a few. This course deals with the foundations and principles of machine and deep learning for network data. Topics include: unsupervised and semi-supervised learning on graphs; graph representation learning; graph signal processing; graph convolutions; graph neural networks; spatiotemporal learning on graphs; scalable algorithms; explainability and privacy of graph neural networks.	
Study Goals	At the end of this course, the student is able to: 1. Recognise key challenges in learning from graph-structured data. 2. Compare representation learning techniques on graphs. 3. Analyse different learning techniques for graph data. 4. Design machine learning solutions for real world graph data tasks	
Education Method	The course consists of: - 12 lectures in which the main concepts are explained and discussed; - 3 hands-on homework labs, in which practical assignments need to be solved individually; - At least two Q&A and feedback sessions about the final project	
Assessment	The final grade of the course consists of the following components: - Oral Exam: Group presentation (weighting 40%) - Group Report: Project report (weighting 60%) Final grade calculation = $0.4 * \text{Oral Exam} + 0.6 * \text{Group Report}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Oral Exam: Resit opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4310	Modeling and Data Analysis and Complex Networks	5
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	At the end of this course, students are able to: 1. Construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies. 2. Model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. 3. Obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: - Assignments (weighting 20%) - Final project (weighting 80%) Final grade calculation = $0.2 * \text{Assignments} + 0.8 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Assignments: re-submit deliverable - Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4315	Performance Analysis	5
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer, electric and transportation networks as well as biological, brain and social networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). In addition, queuing theory is key to assess many kinds of scheduling problems. Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included.	
Study Goals	At the end of this course, the student is able to: 1. Understand and apply mathematical techniques, in particular probabilistic methods and graph theory. 2. To compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) Final grade calculation = $1.0 * \text{Written Exam}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4320	Building Serious Games	5
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 Project-based interdisciplinary course, open to MSc students of all faculties. The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, training, health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose. Assignments for this course will be provided by real-world end-users (e.g. companies or the Science Centre Delft), to whom the group will be reporting throughout the term of the project. Project-based interdisciplinary course, open to MSc students of all faculties (of LDE universities). The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: 1. Identifying and valuing the soft skills necessary to work effectively in interdisciplinary teams 2. Responsibly interacting in a team, integrating its members' talents and expertise in a variety of team roles 3. Adapting with flexibility to the dynamic requirements of a complex external assignment 4. Translating feedback received into proactive personal development steps 5. Formulating research questions, identifying research contributions, and describe them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: 6. Identifying, selecting and deploying the most adequate game technologies for the given serious game domain and constraints 7. Deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small game studio, according to a tight planning. Also a few plenary sessions and/or lectures.	
Assessment	The final grade of the course consists of the following components: - Product grade (weighting 50%). Unique for the whole group, based on both the game itself and the required documentation. - Process grade (individual) (weighting 45%). Including personal contribution, performance, attitude, and peer evaluation. - Final presentation (weighting 5%) The commissioner will be involved both as advisor and as assessor. The final documentation includes writing a scientific paper and preparing it for actual submission to a conference on serious games and/or their application. Final grade calculation = $0.5 \cdot \text{Product grade} + 0.45 \cdot \text{Process grade} + 0.05 \cdot \text{Final presentation}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Final presentation: resit opportunity There is no repair possibility for the other components (product and process), since their assessment necessarily requires an extended development period. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4325	Socio-Cognitive Engineering	5
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic prior knowledge on human-computer interaction is helpful, but not required.	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 Whether you are playing a game in virtual reality, driving a semi-autonomous car, educating yourself with a mobile app, or harmonizing your health and lifestyle with a robot assistant; nowadays intelligent networked information and communication technology is omnipresent. This course focuses on the design of human-aware intelligence into such environments, to support joint human-technology performances that bring about positive human experiences. The focus will be on social robots in general, with practical design and test assignments on robots that support people with dementia, http://rejam.tudelft.nl). In the Socio-Cognitive Engineering (SCE) course (MSc level), you will become acquainted with the application of a coherent set of methods for the design and evaluation of human-agent collaboration. Based on the SCE-method, we will elaborate on the state of the art of intelligent user interfaces (ePartners), such as virtual assistants, robotic teammates, eCoaches, and companion agents. The main topics of study are: - Design methods: Cognitive Engineering, Value Sensitive Design, Scenario-based Design, Claims Analysis, Design Rationale, Design Patterns. - Design for collective intelligence: Knowledge Representation, Ontology Engineering, Mental Models, Theory of Mind, ePartners, Adaptive Automation, Social Robots. - Design Evaluation: Prototyping, Test Methods, Measures, Questionnaires, Ethics. - Human Factors Theories and Models: Human Cognition & Learning, Memory, Emotion, Task Load, Human-Agent Teamwork, Behavior Change and Persuasive Technology.	
Study Goals	At the end of this course, students will be able to: 1. Explain the essential concepts of the design methods addressed in the course. 2. Explain the (dis)advantages of various design methods and their complementarity. 3. Apply the design methods addressed in the course in their research and design projects. 4. Explain what a design rationale is. 5. Construct a design rationale. 6. Create design specifications that are grounded in a design rationale. 7. Evaluate the strengths and weaknesses of a design rationale, e.g. using human-centered evaluations that test the design rationale. 8. Explain some of the state of the art human factors theories, models, and methods relevant to intelligent user interfaces, human-robot collaboration, and ePartner technology. 9. Write a structured report about a design-test cycle, with sufficient detail for a new group of researchers to continue the research. 10. Present work on a design project to an academic audience. 11. Work in a group on collaborative assignments.	
Education Method	LECTURES During the lectures, the teachers will present a range of theories, models, and methods relevant to socio-cognitive engineering. Students are required to read a number of scientific papers which are made available on Brightspace, along with the sheets/slides of the lectures. Together, the sheets/slides and the papers provide the students with the required theoretical knowledge to work on the practical project, and to learn about relevant design methods, human factors theories, conceptual solutions, and design principles. Most of the lectures include practical assignments and discussions stimulating the students to apply the contents of the lecture to their project (also see &#145;Project&#146;). PROJECT In the project, students work in groups to apply the knowledge acquired during the lectures. Students are required to plan, execute, present, and report on a complete design cycle (i.e. design, prototype, and evaluation) for a given design problem. This year (like the past years), the design problem is a social robot for older adults with dementia, and their social environment (https://rejam.tudelft.nl). The objective of the social robot is to improve human&#146;s physical, social, cognitive, and emotional well-being. The students will use the Wiki Socio-Cognitive Engineering (WiSCE) tool to specify the design rationale and its evaluation, step-by-step (see also https://xwiki.ewi.tudelft.nl/). Throughout the course, students will give presentations about their progress, on the design and evaluation of their prototype.	
Literature and Study Materials	Papers from scientific journals on Brightspace. Lecture notes on Brightspace.	
Assessment	The final grade of the course consists of the following components: - Presentations (weighting 15%) - Prototype (weighting 15%) - Project report according to the prescribed format (weighting 60%) - Individual reflection (weighting 10%) Final grade calculation = 0.15 * Presentations + 0.15 * Prototype + 0.6 * Project report + 0.1 * Individual reflection A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Presentations: resit opportunity - Prototype: re-submit deliverable - Project report according to the prescribed format: re-submit deliverable - Individual reflection: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	2024/2025
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Data Science and Artificial Intelligence Technology

Broadening electives

TPM027A	Cyber Risk Management	5
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. A.P. Afghari	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	MOTIVATION: The challenge of selecting the optimal technical, organisational, legal, and other preventive and repressive measures to reduce cyber risks to acceptable levels can only be understood in the context of the application of Cyber Risk Management. Risk Management is about analysing the relationships between threats, incidents and risks (here in the complex world of cyberspace), based on which an adequate set of countermeasures can be designed. SYNOPSIS: Risk (= the potential to losing something of value) can manifest itself in cyberspace in all kinds of ways: values at stake are financial wealth, health, physical condition (of people, materials, goods, infrastructures, etc.), well-being, reputation, privacy, trust, etc. Based on a conceptualisation of cyberspace and its various sub-domains (discussed in the project week of year 1), we introduce risk assessment approaches, both of qualitative and quantitative manner, illustrated with case studies. CONTENT: Cyberspace Risk concepts Probabilistic and adversarial risk System and attacker models Requirements and policies Risk assessment methods Qualitative and quantitative methods Measuring security and effectiveness	
Study Goals	After this course, students are able to: Understand and apply risk concepts to (possible) incidents in cyberspace Understand the theoretical principles of cyber risk management Explain the role of adversarial risk models, system models, and attacker profiles in cyber risk assessment Analyze and apply state-of-the-art cyber risk assessment methodologies to (complex, multi-step) cyber incidents Analyze the (expected) effect of measures that help to prevent the occurrence of cyber incidents Justify investments in cyber security	
Education Method	LECTURES: There are two lecture slots per week EXAM: There is a written exam at the end of the course. LANGUAGE: The course is taught in English. LECTURERS: Prof. dr. ir. Pieter v. Gelder (TUD/TPM) and guest lecturers.	
Assessment	#### Assessment Grading will be based on the individual written exam at the end of the course. Final mark of this course is determined by the grade of the individual exam (100%). The minimum grade for the individual exam to pass this course is 6.0. The grade for the individual exam is 2 years valid. The individual exam is summative. Detailed feedback on the individual exam can be reviewed by students by contacting the teaching staff of this course. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within the study year 2023-2024.	
Elective	Yes	
Targetgroup	MSc	
Category	MSc level	

TPM401A	Technology Entrepreneurship and Innovation	5
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	P.B.M. Vandekerckhove	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	This course offers theoretical fundamentals of entrepreneurship and technology innovation. You create a case study of an existing start-up analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an high-tech start-up to academic research on the topics of technology entrepreneurship and innovation.	
Course Contents	Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, dynamics and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to:	
	LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Describe and apply the high-tech start-up growth process. LO 4 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 5 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 6 Analyse an existing high-tech start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course.	
	Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	Enrollment via Brightspace	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft is needed.	
Category	MSc level	

TPM404B	Technology Entrepreneurship and Internationalization	5
Module Manager	H. Khodaei	
Instructor	J. Schmutzler	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship. Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Identify and analysis international opportunities across different markets. LO 4 Describe and apply the born-global start-up growth process. LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies. LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Elective	Yes	

TPM409A	Digital technology entrepreneurship and management	5
Module Manager	J. Gartner	
Instructor	J. Gartner	
Co-responsible for assignments	H. Khodaei	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Parts	The expected workload is 5 ECTS.	
Summary	This course offers the fundamentals of digital entrepreneurship and the management of emerging (digital) technologies. In this course, participants choose an upcoming digital technology of interest and assess it with different methods from multiple angles to evaluate its ability to create a competitive advantage or disrupt an industry or market. This course helps participants prepare for various career paths, from launching their product/service as an entrepreneur to making informed technology adoption and rejection decisions as an intrapreneur.	
Course Contents	Throughout the course, participants will learn to distinguish between emerging and disruptive technologies, understand influential dynamics such as innovation diffusion, hypes and emotions, and develop the methodical competencies to assess and evaluate an emerging technology in an environment of high information asymmetry. The goal is to learn to leverage the potential of emerging and digital innovations within a specific industry.	
Study Goals	Upon finishing the course, participants will be able to: LO 1 Describe the difference between emerging and disruptive technologies. LO 2 Understand current social and market dynamics that influence the diffusion and success of emerging technologies. LO 3 Develop a critical perspective on conflicting stakeholder interests. LO 4 Analyse an emerging technology's potential for a specific market/industry/company. LO 5 Adapt a business plan to the realities of digital technology. LO 6 Make informed decisions in the context of high information asymmetries.	
Education Method	The education includes: - Lectures - Self-learning study material - Formative Group assignments and hands-on exercises to apply the understanding of relevant material (contributing to 20% of the final grade). - One individual summative assignment to learn how to analyse and create an informed recommendation for action (contributing to 80% of the final grade).	
Assessment	Group work assessments Students will participate in formative group activities throughout the lectures contributing to 20% of the final grade. These activities may include minor in-class and home assignments, presentations, and collaborative projects. Depending on enrollment, group sizes will range from 2 to 6 members. To be considered successful, each assignment requires a grade of 5.0 or above. Individual assessment Students are expected to submit an individual research report at the course's conclusion, comprising 80% of the final grade. This report will be assessed based on the following criteria: - Depth and quality of analysis - Proper utilisation of theoretical concepts - Scope and quality of research and data collection - Strength and coherence of arguments and conclusions - Accuracy and comprehensiveness of the bibliography To pass, the final report must be submitted by November 10th, 2023 latest and achieve a grade of 6.0 or above. It is not possible to repeat or resubmit the report.	
Enrolment / Application	This course is part of the MOT specialisation ICT Design and Management and is also open for students from across the Delft University of Technology faculties. You can directly enrol into Brightspace. Brightspace enrollment will be activated by education support a couple of weeks before the course starts. No other means of registration, such as the app MyTuDelft, are needed.	
Special Information	Students are asked to register/enroll on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

TPM411A	Idea to Start-up IT & AI	5
Module Manager	E. Çelik	
Instructor	Dr. A. Giga	
Instructor	E. Çelik	
Responsible for assignments	E. Çelik	
Co-responsible for assignments	Dr. A. Giga	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity using emerging digital technologies. Applications can range from mobility, cyber security to predictive maintenance etc. You will test its validity, and build a viable business model a roadmap from an idea to an innovation-driven company. On this journey, you will be coached by lecturers and learn practical insights from guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one.	
Course Contents	You will focus on information and computing technologies (e.g AI, big data, blockchain) that are transforming traditional industries and societies at large. To analyse the opportunities for new startups we focus on entrepreneurial tools such as the opportunity canvas; value proposition development and customer empathy mapping, market segmentation and value chain analyses and the initial operations to develop a minimum viable product.	
Study Goals	Develop a business Idea by identifying a relevant economic problem or unmet need that is addressed by digital solutions, and advanced information and computing technologies. Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment Analyse the competitive landscape and devise your products competitive advantage, given specific challenges of protecting IP in the digital world. Create a blueprint of your product. Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. Understand how these strategies adjust when trying to implement your solution in well-established systems of traditional industries. Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	You will work in teams which are formed in the first sessions of the course. One aspect of a startup will be discussed each week and you will apply into your own projece immediately. You will present your work, provide peer-feedback as well and receive advise from coaches. The course ends with a jury pitch and a 4-pager of the business proposition. Team coaching will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Books	The primary textbook for this course is an organic handbook by Bill Aulet: Disciplined Entrepreneurship: 24 Steps to a Successful Startup. Wiley, 2023.	
Assessment	The assessment is based on your project. It includes class assignments and participation (40%), final business model report (30%), online pitch presentation (15%), and team evaluation (15%). Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

Year	2024/2025
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Data Science and Artificial Intelligence Technology

Thesis Project 2024

DSAIT5000	Thesis Project	45
Responsible Instructor	Dr.ir. W.P. Brinkman	
Responsible Instructor	Dr. R. Hai	
Instructor	Dr. P.K. Murukannaiah	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The Data Science and Artificial Intelligence Technology (DSAIT) Thesis Project is the final part of the Master's degree programme. During this project, the student will be required to demonstrate their ability to conduct a DSAIT research project, design a solution, communicate and reflect about it. Students can use ProjectForum.tudelft.nl to orientate themselves on possible thesis project topics, and contact potential thesis advisors and possibly a daily (co-) supervisor. When the student and thesis advisor agree on a thesis project topic, they plan a kick-off meeting (start date of the project); here, they check the suitability of the topic and discuss planning, milestones of the project and the resources and facilities offered by the faculty. The student and thesis advisor/supervisor plan two formative assessment moments, where they receive feedback from their thesis advisor/supervisor using a standardized rubric. The first formative feedback moment, the first stage review, is 10 working weeks after the kick-off meeting. The second formative feedback moment, the greenlight review, takes place 14 working weeks after the first stage review. After a positive greenlight review, the student plans a thesis defence with the thesis committee after 6 working weeks.	
Study Goals	It is possible for students to do their thesis project with an external company, under supervision of a thesis advisor. At the end of this course, the student is able to: 1. Conduct a research project in a DSAIT field. 2. Design DSAIT solutions. 3. Communicate about their research. 4. Reflect on their research and the DSAIT field in the temporal and societal context.	
Education Method	Project under supervision of thesis advisor and/or daily (co-)supervisor	
Literature and Study Materials	Project Forum, MaRe, Brightspace, student manual	
Assessment	At the end of the project, the student submits their Masters thesis, written in English, and gives an oral presentation of their work to the thesis committee, followed by a question-and-answer session. The thesis committee will announce the final mark, which is based on the standardized rubric.	

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